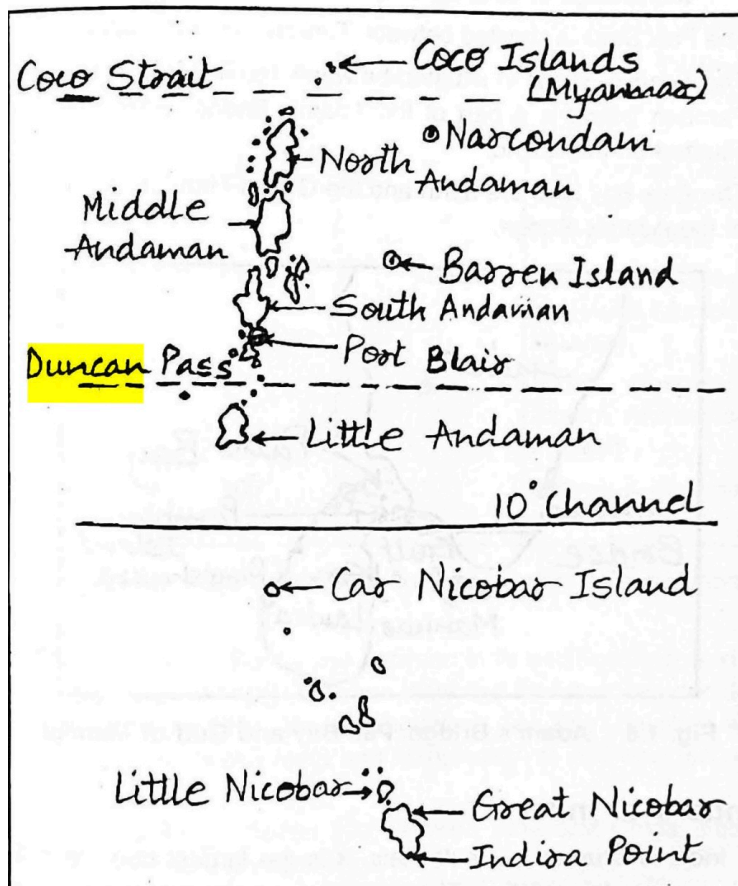


ANSWERS & EXPLANATIONS
GENERAL STUDIES (P) TEST – 6334

- Q 1. C** • **The purpose of Article 32** is to provide a guaranteed, effective, expeditious, inexpensive, and summary remedy for protection of the fundamental rights. Only the Fundamental Rights guaranteed by the Constitution are enforced under Article 32 and not any other rights like non-fundamental constitutional rights, statutory rights, customary rights, and so on. In other words, the Supreme Court under Article 32, cannot determine a question that does not involve Fundamental Rights. **Article 32 cannot be invoked simply to determine the constitutionality of an executive order or legislation unless it directly infringes any of the fundamental rights. Hence, statement 1 is correct.**
- The Supreme Court has held that while the **Parliament has the power to amend the Constitution, it cannot destroy or abrogate its basic structure. The basic structure doctrine, established by the Supreme Court in the landmark case of Kesavananda Bharati v. State of Kerala, ensures that certain fundamental features of the Constitution, including the right to constitutional remedies, cannot be abridged or eliminated through constitutional amendments. Hence, statement 2 is correct.**
- Q 2. A** • **A credit rating agency** is an entity which assesses the ability and willingness of the issuer company for timely payment of interest and principal on a debt instrument. **Credit rating is an assessment of the probability of default on payment of interest and principal on a debt instrument.** It is not a recommendation to buy, sell or hold a debt instrument. In the Indian context, debt instruments rated 'BBB-' and above are classified as investment grade ratings. Instruments that are rated 'BB' and below are classified as speculative grade category ratings in which case the ability to meet the payment obligations is considered to be "speculative". **In India, the issuer company pays for the credit rating. Hence, statement 2 is correct.**
- **Credit rating agencies are regulated by SEBI.** The SEBI (Credit Rating Agencies) Regulations, 1999 govern the credit rating agencies and provide for eligibility criteria for registration of credit rating agencies, monitoring and review of ratings, requirements for a proper rating process, avoidance of conflict of interest and inspection of rating agencies by SEBI, amongst other things. **RBI accredits CRAs for credit ratings of bank loans / facilities,** authorizes CRAs for independent credit evaluation of resolution plans and approves CRAs for rating of public deposits raised by NBFCs.
 - **The Reserve Bank of India and SEBI have made credit rating compulsory in respect of all non-government debt securities where the maturities exceed 18 months.** Credit rating has also been made compulsory for commercial papers. As per Reserve Bank of India guidelines rating of P2 by CRISIL or A2 by ICRA or PP2 by CARE is necessary for commercial papers. **Hence, statement 3 is not correct.**
 - **A credit rating agency provides an opinion relating to future debt repayments by borrowers. A credit bureau provides information on past debt repayments by borrowers. Hence, statement 1 is correct.**

- Q 3. D** • **Barren Island is an uninhabited island in the Andaman Sea.** It is the only confirmed active volcano in the Indian subcontinent. It is a part of the Indian Union territory of the Andaman and Nicobar Islands.
- **Narcondam, India's easternmost island,** is a small volcanic island located in the northern Andaman Sea. The island's peak rises to 710 m above mean sea level, and it is formed of andesite.
 - **Duncan Passage is a strait in the Bay of Bengal.** It is about 48 km wide; it separates Rutland Island (part of the Great Andaman and South Andaman administrative division) to the north, and Little Andaman to the south. West of Duncan Passage is the Bay of Bengal; east is the Andaman Sea.
 - **The Coco Strait,** more commonly known as the Coco Channel, is a water body that separates the North Andaman Islands of India from Myanmar's Coco Islands. This strategically important channel lies in the northeastern Bay of Bengal and is a significant area for geopolitical reasons between India and Myanmar.



- Hence, option (d) is the correct answer.

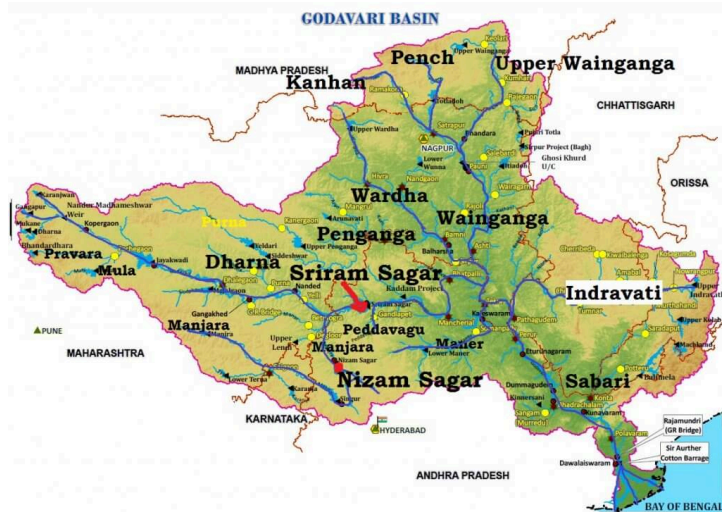
- Q 4. A** • For aquatic organisms the quality (chemical composition, pH) of water becomes important. The salt concentration (measured as salinity in parts per thousand), is less than 5 in inland waters, 30-35 in the sea and > 100 in some hypersaline lagoons.
- Some organisms are tolerant of a wide range of salinities (euryhaline) but others are restricted to a narrow range (stenohaline). Many freshwater animals cannot live for long in seawater and vice versa because of the osmotic problems, they would face.
 - **Euryhaline organisms are able to adapt to a wide range of salinities.** An example of a euryhaline fish is the molly which can live in freshwater, brackish water, or saltwater. The green crab is an

example of a euryhaline invertebrate that can live in salt and brackish water. **Hence option (a) is the correct answer.**

- **Stenohaline** describes an organism, usually fish, that cannot tolerate a wide fluctuation in the salinity of the water.

- Q 5. C**
- The history of Vijayanagar Empire constitutes an important chapter in the history of India. Four dynasties – Sangama, Saluva, Tuluva and Aravidu – ruled Vijayanagar from A.D. 1336 to 1672.
 - The sources for the study of Vijayanagar are varied such as literary, archaeological and numismatics.
 - Many **foreign travellers** visited the Vijayanagar Empire and their accounts are also valuable.
 - The Moroccan traveller, Ibn Battuta, Venetian traveller Nicolo de Conti, Persian traveller Abdur Razzaq and the Portuguese traveller Domingo Paes were among those who left valuable accounts of the socio-economic conditions of the Vijayanagar Empire.
 - **Duarte Barbosa (Portuguese):** He had given a vivid account of Vijaynagara governance under Krishna Deva Raya in his book, "An Account of Countries Bordering the Indian Ocean and their Inhabitants". He provided insights into the local customs, including marriage ceremonies and social gatherings.
 - According to **Domingo Paes**, "Krishnadeva Raya was the most feared and perfect king there could possibly be." Paes mentioned the beautiful houses of the rich and the large number of their household servants. He also referred to the flourishing devadasi system.
 - **Nicolo Conti**, who visited Vijayanagara in 1420 referred to the prevalence of slavery. He also gave a detailed account of the layout of the city of Vijayanagara.
 - The Persian traveller **Abdur Razzaq** visited Vijayanagara during the reign of Deva Raya II. He gave a glowing account, mentioning that the country was well cultivated and that the troops numbered about eleven lakhs.

- Q 6. C**
- Left Bank: Dharna, Penganga, Wainganga, Wardha, Pranhita, Indravati, Sabari, Purna.
 - Right Bank: Pravara, Mula, Manjra, Peddavagu.
 - **Penganga:** It originates in the Ajanta ranges and is a major part of the northern drainage system. It eventually joins the Wardha (which forms the Pranhita). It is classified as a Left-bank tributary.
 - **Indravati River:** Originating in the Eastern Ghats (Dandakaranya), it flows westward to join the Godavari. It is one of the most significant Left-bank tributaries.
 - **Purna:** This river flows through the Purna valley of Maharashtra and joins the Godavari from the North. It is a Left-bank tributary. (Note: Not to be confused with the Purna river that flows into the Tapti).



- Hence, option (c) is the correct answer.

Q 7. D Mahalwari System:

- A modified version of the zamindari settlement, **introduced in the Gangetic valley, the North-West Provinces, parts of Central India, and Punjab**, was known as the Mahalwari System.
- **The revenue settlement was to be made village by village or estate (mahal) by the estate with landlords or heads of families who collectively claimed to be the landlords of the village or the estate.**
- In Punjab, a modified Mahalwari System known as the village system was introduced.
- **In Mahalwari areas also, the land revenue was periodically revised. Hence all the statements are correct.**

Q 8. C • Cinnabar is the common name for mercury(II) sulfide (HgS). It is a heavy, bright scarlet to brick-red mineral and serves as the primary and most important ore for the extraction of elemental mercury (quicksilver). **Hence, option (c) is the correct answer.**

- **Formation & Occurrence:** It is typically found as a vein-filling mineral in regions associated with recent volcanic activity and alkaline hot springs. Major historical deposits include the Almadén mine in Spain and the Huancavelica mine in Peru.
- **Historical Use:** Due to its vivid color, cinnabar was used for thousands of years as a pigment known as vermilion. However, its use has largely been discontinued due to the high toxicity of mercury.

Q 9. A • At the time of India's independence in 1947, the Indian subcontinent was a patchwork of British-controlled territories and 562 princely states, which were autonomous regions ruled by local monarchs under the suzerainty of the British Crown. The integration of these princely states into the newly independent Indian Union was a complex and critical task, crucial for the political and territorial consolidation of India. a few important Princely States and Their Unification:

Hyderabad: Ruled by Nizam Osman Ali Khan. The Nizam of Hyderabad initially sought independence and refused to accede to India.

- **After a prolonged stand-off and growing internal unrest, the Indian government launched "Operation Polo" in September 1948, a police action that led to the incorporation of Hyderabad into India.**

Kashmir: Maharaja Hari Singh of Kashmir hesitated between joining India or Pakistan, leading to a standstill agreement. However, following a tribal invasion from Pakistan, he chose to accede to India in October 1947, seeking military assistance.

- The Instrument of Accession was signed, but the region remains a point of contention between India and Pakistan.

Junagadh: The Nawab of Junagadh opted to join Pakistan, despite the state being predominantly Hindu.

- This decision led to civil unrest. **The Indian government intervened, and after a plebiscite in February 1948, where the majority voted to join India, Junagadh was integrated into the Indian Union.**

Travancore: The Maharaja of Travancore declared his intention to remain independent.

- Following negotiations and pressure from Indian leaders, Travancore acceded to India on July 1, 1949.
- **Mysore:** Maharaja Jayachamarajendra Wadiyar signed the Instrument of Accession on August 9, 1947, integrating Mysore into the Indian Union smoothly.

Manipur: Manipur had its own constitution and elected assembly and was initially hesitant to join India.

- **After intense negotiations and a brief period of political unrest, Maharaja Bodhchandra Singh signed the Instrument of Accession in September 1949, leading to Manipur's integration into India.**
- **Hence option (a) is the correct answer.**

Q 10. D 7 projects approved under Electronics Component Manufacturing Scheme (ECMS)

- Projects will enable large-scale production of components used across smartphones, automobiles, medical devices, telecommunications, and industrial systems.
- **ECMS was notified in April 2025 with a total outlay of ₹22,919 crore.**

About ECMS

- Objective: Build a strong and self-sustaining ecosystem for electronics component manufacturing in the country.

Nodal Ministry: Ministry of Electronics and Information Technology (MeiTY)

- **Targeted Segments:** It includes sub-assemblies (Display and camera module), Bare components like Non-Surface Mount Devices, etc.
- Tenure: Turnover Linked Incentive: 6 years with 1 year of gestation period
- Capex Incentive: 5 year

India's Electronics Sector

- **It contributes around 3.4% of the country's GDP. Hence statement 1 is not correct.**
- Electronics production has grown from ₹1.9 lakh crore in 2014–15 to ₹11.3 lakh crore in 2024–25, marking a six-fold increase.
- Exports rose from ₹38,000 crore to ₹3.27 lakh crore in the same period, reflecting an eight-fold jump.

- **Electronics have emerged as India's third largest and fastest growing export category in 2024–25 (rising from the seventh position in 2021–22).**
- India is now the world's second largest mobile phone manufacturer. China is the world's largest manufacturer of mobile phones, contributing approximately 67% to global production. It dominates the industry, producing smartphones for major brands like Huawei, Xiaomi, Oppo, Vivo, Honor, and Lenovo. **Hence statement 2 is not correct.**

- Q 11. C**
- In the 18th century, three Carnatic Wars were fought between various Indian rulers and British and French East India Company on either side. **These wars resulted in the establishment of the political supremacy of British East India Company.** The French company was reduced in the areas around Pondicherry only. The commercial **and maritime rivalry between France and England was the primary reason behind these wars**
 - **First Carnatic War (1746-1748) :** English navy under Barnett captured some French ships. The French governor of Pondicherry, Dupleix attacked the English in retaliation in 1746 and this led to the beginning of first Carnatic War. **Treaty of Aix-La-Chappelle brought an end to the first round of Anglo-French conflicts in India as well.**
 - **Second Carnatic War (1749-1754):** Anglo-French rivalry, continued in India although it had ended in Europe. In 1748, Nizam of Hyderabad Nizam-ul-Mulk Asaf Jah died, which resulted in a war of succession. Muzaffar Jang, who aspired to become the Nizam of Hyderabad and Chanda Sahib, a candidate for the throne of Arcot was supported by French Governor. After Victory in Battle of Ambur in 1749, Muzaffar Jung became the Nizam and Chanda Sahib the Nawab of Muhammad Ali, (son of Anwar Uddin) who was supported by British escaped to Tiruchirappalli. **In 1751 the British commander Robert Clive captured Arcot, i.e. the capital of the Carnatic.** Chanda Sahib was treacherously murdered by the Raja of Tanjore. **Later, Duplex was recalled. The war concluded by the Treaty of Pondicherry in 1755.** According to this treaty each party was left in possession of the territories that it occupied at the time of the treaty. **Hence, unlike other 2 wars, it was not influenced by outside factors.**
 - **Third Carnatic War (1758-1763): The outbreak of the Seven Years War (1756-1763) in Europe was the cause of the Third Carnatic War (1758-1763).** The British General Sir Eyre Coote defeated, Count de Lally (the commander of the French troops) at Wandiwash in 1760. Battle of Wandiwash ended almost a century of conflict over supremacy in India and availed the British East India company a far superior position in India compared to the other European traders. **The Seven Years War concluded by the Treaty of Paris in 1763 and this also led to the ending of Third Carnatic War.** The French got Pondicherry, Karaikkal, Mahe and Yenam but condition applied was these were to be never fortified.
 - **Hence, both statements are correct.**

- Q 12. C**
- **IUCN World Heritage Outlook 4 report categorises Western Ghats as being of “significant concern”.**
 - **Apart from Western Ghats, two national parks in India – Assam’s Manas national park and West Bengal’s Sundarbans national park – have also been categorised as being of “significant concern”.**

IUCN World Heritage Outlook assesses the conservation prospects of all World Heritage sites inscribed for their natural values.

- The outlook categorizes sites in **four categories - Good, Good with some concerns, Significant Concern, and Critical.**
- **Significant concern outlook** means that site's values and underlying attributes are threatened by several current and/or potential threats, requiring additional conservation measures.

About Western Ghats

- Length: About 1,600 km long, they are a chain of mountains older than the Himalayas and running parallel to India's western coast.
- **Spread across six states: Gujarat, Maharashtra, Goa, Karnataka, Kerala and Tamil Nadu. Andhra Pradesh is not included in it.**
- **Hence option (c) is the correct answer.**

Q 13. D • Healthy ecosystems are the base for a wide range of economic, environmental and aesthetic goods and services. The products of ecosystem processes are named as ecosystem services, for example, healthy forest ecosystems purify air and water, mitigate droughts and floods, **cycle nutrients, generate fertile soils, provide wildlife habitat**, maintain biodiversity, **regulates climate**, pollinate crops, provide storage site for carbon and also provide aesthetic, cultural and spiritual values.

Q 14. C • **Madhavacharya** in the 13th century propounded the **Dvaita philosophy** and **opposed the Shankaracharya philosophy of Advaita** and **according to Dvaita philosophy, Brahman, the self and the world are completely distinct.**

• **His Sarva Darshan Sangrah is a philosophical work that analysed and explained the system of Indian philosophy.** Based on the Upanishads, his theory postulates that the individual soul is created by God. **Hence, option (c) is the correct answer.**

Q 15. B • Recent Context: The President approved the conferment of 131 Padma Awards. Awardees include 19 women, 6 persons from Foreigners/NRI/PIO/OCI category, and 16 Posthumous awardees. There are 2 Duo cases (counted as one).

- According to the selection process, the **Padma Awards Committee is headed by the Cabinet Secretary**, not the Prime Minister. While the Prime Minister constitutes the committee annually and approves the final list, he does not chair the committee himself. **Hence, statement 1 is not correct.**
- The criteria explicitly state that Government servants, including those working with PSUs, are not eligible, except for those in the categories of doctors and scientists. **Hence, statement 2 is correct.**
- In the case of *Balaji Raghavan vs. Union of India* (1996), the Supreme Court of India ruled that Padma Awards are not titles under Article 18(1) of the Constitution and do not violate the principle of equality. As non-hereditary, honorary recognitions of merit, they are not to be used as prefixes or suffixes to a recipient's name. **Hence, statement 3 is not correct.**

Q 16. A • **World Economic Forum (WEF):** The WEF publishes the Global Cybersecurity Outlook annually. The 2026 report specifically emphasizes the "Cyber-Equivocation" gap, where smaller organizations are falling behind larger ones due to the rapid advancement of Generative AI and the rising cost of cyber-insurance. **Hence, option (a) is the correct answer.**

- **International Telecommunication Union (ITU):** While the ITU is a major player in this space, it releases the **Global Cybersecurity Index (GCI)**, which ranks countries based on their legal,

technical, and organizational commitments to cybersecurity.

- Interpol & UNODC: These organizations focus more on the enforcement and legal frameworks for tackling cybercrime rather than global outlook reports on corporate and systemic resilience.

- Q 17. B**
- **Article 80** provides that the Council of States shall consist of twelve members to be nominated by the President in accordance with the provisions of clause and not more than two hundred and thirty-eight representatives of the States and of the Union territories. **Article 330** provides for reservation of SC/STs in Lok Sabha but **doesn't provide any reservation in Rajya Sabha**.
 - **Article 324** of the Constitution provides that the power of superintendence, direction and control of elections to parliament (**Both Rajya Sabha and Lok Sabha**), state legislatures, the office of president of India and the office of vice - president of India shall be vested in the election commission. **Hence the statement 1 is not correct.**

The following provisions were borrowed from the **South African Constitution**

- Procedure for amendment of the Constitution
- **Election to the Rajya Sabha. Hence statement 2 is correct.**

- Q 18. B**
- The Sikhs in the 18th century were organized into 12 misls or confederacies, which operated in different parts of the province.
 - These misls fully cooperated with one another.
 - They were originally based on the principle of equality with all members having an equal voice in deciding the affairs of the misl and in electing its chief and other officers. However, gradually the democratic and plebeian character of the misls disappeared and powerful feudal chiefs and zamindars dominated them. Ranjit Singh, who rose to prominence at the end of the 18th century, was the chief of Sukerchakia Misl.

- Q 19. D**
- The **Kanpur Conspiracy Case** was one of the major **anti-revolutionary actions** undertaken by the colonial government in Indian history. The political activists were arrested from **Kanpur Nagar, Uttar Pradesh** and hence the name.

Background:

- The **Non-Cooperation Movement** was **called off** by Gandhiji due to the Chauri Chaura incident.
- **Kanpur** emerged as one of the key hubs of **revolutionary activities** in North India, the aim of the same was: use of **violent means** to overthrow imperialism once and for all.

The **Kanpur Conspiracy Case** was thus against the **emerging communists**, who were absolutely **hated by the British Government**. In this row, certain people were caught by the Government and were **tried for conspiring against the Government**. The **Charge** against them was “to **deprive the King Emperor of his sovereignty of British India**, by the complete separation of India from imperialistic Britain by a violent revolution. The names of these people included:

- **M N Roy,**
- **Muzaffar Ahamed,**
- **S A Dange,**
- **Shaukat Usmani,**
- **Nalini Gupta,**

- Singaravelu Chettiar, and
- Ghulam Hussain.
- Through these trials, the **government aimed to crush the Socialist movement** in India entirely. However, this effort ended up backfiring, and the **leftist movement** became popular with the attention of young activists.

Q 20. B • Recent Context: The Prime Minister of Vietnam undertook a State visit to India.

Initiatives taken by India to strengthen relations with Vietnam:

- **Mekong-Ganga Cooperation (MGC):** An initiative by India and five ASEAN countries, (Cambodia, Lao PDR, Myanmar, Thailand, and Vietnam) for cooperation in tourism, culture, education, as well as transport and communications. Hence, option (b) is the correct answer.
- Quick Impact Projects: It is being undertaken by India in various provinces of Vietnam under the MGC framework
- India provides training, capacity building and educational courses in India for Vietnamese nationals.
- India handed over an indigenously built missile corvette INS KIRPAN to Vietnam in 2023.

Q 21. B • Statement 1 is not correct: The office of 'whip' is mentioned neither in the Constitution of India nor in the Rules of the House nor in a Parliamentary Statute. It is based on the conventions of the parliamentary government. Every political party, whether ruling or Opposition has its own whip in the Parliament. He is appointed by the political party to serve as an assistant floor leader.

- **Statement 2 is correct:** He is charged with the responsibility of ensuring the attendance of his party members in large numbers and securing their support in favour of or against a particular issue. He regulates and monitors their behaviour in the Parliament. The members are supposed to follow the directives given by the whip. Otherwise, disciplinary action can be taken.

Q 22. A • The SARFAESI Act, or Securitisation and Reconstruction of Financial Assets and Enforcement of Security Interest Act 2002 allows banks and financial institutions to auction residential and commercial assets of defaulters to recover their loans without the need for lengthy court procedures or prior judicial approval. **Hence, statement 1 is correct.**

- The Act is not applicable for loans where the outstanding amount is less than ₹1 lakh. **Hence, statement 2 is correct.**
- Cooperative banks are covered under the SARFAESI Act. **Hence, statement 3 is not correct.**

Q 23. C About the Indus Water Treaty:

- **Genesis:** Signed in 1960 between India and Pakistan mediated by the World Bank.

Water usage rights:

- **Eastern Rivers (Ravi, Beas, and Sutlej) for India's unrestricted use.**
- Western Rivers (Indus, Jhelum, and Chenab) are allocated to Pakistan, with India allowed for specific non-consumptive uses like navigation, floating of timber or other property, flood protection or flood control, fishing or fish culture. **Hence, option (c) is the correct answer.**

- Q 24. A** • A harmful way in which acids can enter the lakes is spring acid shock. When the snow melts in spring rapidly due to a sudden temperature change, the acids and chemicals in the snow are released into the soils. The melted snow then runs off to streams and rivers, and gradually make their way into the lakes. **The introduction of these acids and chemicals into the lakes causes a sudden drastic change in the pH of the lakes - hence the term "spring shock or spring acid shock".**
- The aquatic ecosystem has no time to adjust to the sudden change. In addition, springtime is an especially vulnerable time for many aquatic species since this is the time for reproduction for amphibians, fish and insects. Many of these species lay their eggs in the water to hatch.
 - The sudden pH change is dangerous because the acids can cause serious deformities in their young or even annihilate the whole species since the young of many of such species spend a significant part of their life cycle in the water. **Hence option (a) is the correct answer.**

- Q 25. C** • **Inter-Tropical Convergence Zone (ITCZ)**
- The Inter-Tropical Convergence Zone (ITCZ) is a near-equatorial belt where the trade winds from both hemispheres converge. Intense solar heating makes surface air warm and moist, promoting strong convection and frequent thunderstorms.
 - **Convergence** of northeast and southeast trades.
 - **Vertical uplift** leads to cloud formation and heavy precipitation.
 - Distinct from **subtropical highs** (subsidence) and **subpolar lows** (mid-latitude interactions).
 - Subtropical highs are zones of subsiding dry air around 30°, not convergence and uplift.
 - The monsoon trough is a seasonal low-pressure extension over South Asia; the broader global convergence zone is the ITCZ. Whereas, Subpolar lows occur around 60° where westerlies and polar easterlies interact, not where trade winds converge.
 - **Hence option (c) is the correct answer.**

- Q 26. B** • **The goal of Atal Bhujal Yojana (Atal Jal) is to demonstrate community-led sustainable groundwater management which can be taken to scale. The major objective of the Scheme is to improve the management of groundwater resources in select water-stressed areas in identified states viz. Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan, and Uttar Pradesh. Hence, statement 1 is correct but statement 2 is not correct.**

Facts of the scheme:

- **Type: Central Sector Scheme**
- **Funding:** 50% through World Bank loan and the remaining through budgetary support
- **Women empowerment:** Mandatory participation of women
- **Incentives to States:** States receive incentives under the scheme based on performance and achievements of predefined indicators.
- **Grassroots level governance:** Active participation of the communities. Gram Panchayat wise, Water Security Plans (WSPs) & their implementation.
- **IEC activities:** For sustainable groundwater management. Institutional Strengthening and Capacity Building.
- **Nodal Implementing agency:** Central Ground Water Board (CGWB) of the Department of Water Resources, River Development and Ganga Rejuvenation. Hence, statement 3 is

correct.

- Q 27. B**
- The Green Credit Programme (GCP) is an innovative, market-based mechanism designed to incentivise voluntary environmental actions across various sectors.
 - The programme provides tradable 'green credits' for a range of voluntary environmental activities beyond just carbon, such as water conservation, afforestation, sustainable agriculture, and waste management. **Hence, statement 1 is correct.**
 - The GCP aims to complement existing regulatory frameworks by offering a market-based incentive, encouraging participation from individuals, industries, and organisations in pro-environmental behaviour. **Hence, statement 2 is correct.**
 - Green credits are distinct from carbon credits. They cannot be used to meet mandatory carbon offset obligations for international markets. Instead, they can be traded on a domestic platform and used by companies to meet their environmental, social, and governance (ESG) mandates or other voluntary environmental targets. **Hence, statement 3 is not correct.**
- Q 28. A**
- The depressions that form in the monsoon season are called the monsoon depressions. These are low-pressure areas with two or three closed isobars (at 2 hPa interval), which cause most of the monsoon rains. These can be of Bay origin, Land origin or Arabian Sea origin. Their shape is roughly elliptical and its horizontal extension is about 1000's of Kms of surface. Its vertical extension is about 6-9 kms.
 - **Monsoon depression is cold core system (central temperature colder than the environment) over the surface and in the lower levels and warm core in upper levels (central temperature warmer than environment). The Maximum wind strength and intensity can be noticed at the levels of 0.9km or 1.5 km. The monsoon depressions tilt southwards with height and if monsoon depression is moving westward, the heavy rainfall is mainly concentrated in the SW quadrant. Hence, statement 1 is correct.**
 - **Due to the high vertical wind shear present during South west monsoon season, monsoon depressions are generally do not intensify into cyclonic storms. Hence, statement 2 is not correct.**
 - The depressions forming in the pre-monsoon season and post-monsoon season intensify into a cyclonic storm. The average diameter of Post monsoon storms is about 1200 km, whereas in the pre-monsoon season it is about 800 km, however, intensity does not depend on size. The cyclonic storm is a warm core phenomenon where the temperatures at the centre are warmer than the surrounding (areas) regions. The maximum warming occurs at the 300 hPa level.
- Q 29. B**
- Royal Commission on Indian Currency and Finance was setup in 1925. Its chairman was Edward Hilton Young. It was setup to examine and report on the Indian Exchange and Currency system and practice, to consider whether any modifications are desirable in the interests of India, and to make recommendations. Some of its recommendations were:
- It recommended the setting up of an institution the Reserve Bank of India - which was to be entrusted with pure central banking functions. It was to replace the Imperial Bank and Imperial Bank to do only commercial banking business.
 - The RBI was set up on the basis of the recommendations of the commission.

- The commission also recommended a full-fledged gold bullion standard. It recommended that the Indian Currency must be linked with gold in a manner that is real and conspicuously visible.
- **The Hilton Young Commission (1925) recommendations were accepted by the Government. Hence option (b) is correct.**
- The Muddiman Committee or the Reforms Enquiry Committee (1924) was a committee lead by Sir Alexander Muddiman, organized by the British and Indian government, to meet the demand of Indian leaders in the context of Indians new Purna Swaraj declaration (India's Independence). This committee would aid in investigating the diarchy issue on the Constitution as set up in 1921 under the Indian Council Act of 1919.
- Butler Committee was set up in 1927 to examine the nature of the relationship between the states and government.
- Hartog committee was appointed by the British Indian government to survey on the growth of education in India. The Hartog committee on education submitted its report in 1929. It had devoted more attention to mass education than the secondary and University education.

Q 30. D • Balance of Payments (BoP) statistics systematically summaries the economic transactions of an economy with the rest of the World (i.e.transactions between resident & non resident entities) during a given period. It comprises of current and capital & financial accounts. It comprises of current and capital & financial accounts.

- **Current Account:** Current account covers all transactions (other than those in financial items) that involve economic values and occur between resident and non resident entities. Also covered are offsets to current economic values provided or acquired without a quid pro quo. Specifically, the major classifications are goods and services, income, and current transfers. Thus, current account is a broader measure than trade balance as it also includes income(investment income & compensation of employees) & current transfers.
- The current account of the BoP provides information not only on international trade in goods, but also on international transactions in services. Current account of the BoP, transactions can be classified into merchandise (exports and imports) and invisibles.

Invisible transactions are further classified into three categories, namely

- Services- **travel**, transportation, insurance, Government not included elsewhere (GNIE) and miscellaneous (such as, communication, construction, financial, software, news agency, **royalties**, management and business services etc);

Income (investment income & compensation of employees); and

- **Investment income is money received in interest payments, dividends, capital gains** realized with the sale of stock or other assets. It consists of **direct investment income, portfolio investment income**, and other investment income.
- **Current Transfers** (grants, gifts, **remittances**, etc.) which do not have any quid pro quo. **Hence, option (d) is the correct answer.**
- **Capital Account:** The major components of the capital account are (a) capital transfers and (b) acquisition/disposal of nonproduced, nonfinancial assets.

Q 31. D • “Scheme of Fund for Regeneration of Traditional Industries (SFURTI)” is a programme of the Ministry of Micro Small and Medium Enterprises, Government of India and was launched in the year 2005.

- **Aim:** Organize the traditional industries and artisans and producers into collectives and support them for quality and competitive products for long-term sustainability of the sector and its artisans.
- **Nodal Implementing Agency:** KVIC for Khadi and Village Industry clusters and Coir Board for Coir based clusters.
- **Implementing Agencies (IAs):** NGOs, institutions of the Central and State Governments and semi-Government institutions, Panchayati Raj institutions (PRIs) registered Producer Collectives, etc.
- **Focus:** physical infrastructure creation, technology upgradation, training, product development, innovation, etc.
- **Participation of Private entities:** Corporates and Corporate Social Responsibility (CSR) foundations with expertise in cluster development will be encouraged to participate as IAs.
- **Formation of SPV:** Formation of a SPV dedicated for the purposes operating the SFURTI Cluster is mandatory

Type of clusters	Per cluster budget limit
Regular clusters (upto 500 artisans)	Rupees 2.5 Crore
Major clusters (more than 500 artisans)	Rupees 5 Crore

- Hence option (d) is the correct answer.

Q 32. D Article 214 of the Constitution of India states that:

- **There shall be a High Court for each State. Hence Statement-II is correct.**
- For the purposes of this Constitution, the High Court exercising jurisdiction in relation to any Province immediately before the commencement of this Constitution shall be deemed to be the High Court for the corresponding State.
- The **Seventh Amendment Act of 1956 authorized the Parliament to establish a common high court** for two or more states or for two or more states and a union territory. **Hence Statement-I is not correct.**
- The territorial jurisdiction of a high court is co-terminus with the territory of a state. Similarly, the territorial jurisdiction of a common high court is co-terminus with the territories of the concerned states and union territories.
- At present, there are 25 high courts in the country. Among the union territories, Delhi has a separate high court, and the union territories of Jammu and Kashmir and Ladakh have a common high court.
- The other union territories fall under the jurisdiction of different state high courts. The Parliament can extend the jurisdiction of a high court to any union territory or exclude the jurisdiction of a high court from any union territory.
- Hence option (d) is the correct answer.

Q 33. B

The Constitution provides for the creation and abolition of legislative councils in states. **Under Article 169 of the Indian constitution, Parliament may create or abolish the Council in a state if the Legislative Assembly of that state passes a resolution to that effect by a special majority. Hence statement 1 is not correct.**

- The resolution for the creation of legislative council must be passed by a special majority, i.e., a **majority of total membership of assembly and a majority of not less than two-thirds of the assembly present and voting.**
- The **Vice President can be removed (Article 67(b)) by a resolution passed by the Rajya Sabha by an effective majority (refers to a majority of the total membership of the House excluding the vacant seats).** Then it must be agreed to by the Lok Sabha by a simple majority. **Hence statement 2 is correct.**
- **Absolute Majority refers to more than half of the total number of members of the house, whether they are present or not. Hence statement 3 is not correct.**

- Q 34. D**
- BOLD-QIT is a project aimed at installing technical systems under the Comprehensive Integrated Border Management System (CIBMS). It was specifically launched to secure the highly porous and riverine borders where physical fencing is not feasible.
 - The project was first implemented by the Border Security Force (BSF) in the Dhubri district of Assam along the India-Bangladesh border. This area is dominated by the Brahmaputra River and its numerous tributaries. It uses a "virtual fence" consisting of microwave communication, data networks, thermal imagers, and various sensors. These tools provide real-time feeds to BSF control rooms, allowing Quick Reaction Teams (QRTs) to intercept illegal crossings and smuggling activities instantly.
 - It reduces the need for continuous physical patrolling in difficult terrain and provides 24/7 surveillance regardless of weather conditions.
 - **Hence, option (d) is the correct answer.**

- Q 35. C**
- National Conference of Chairpersons of State Public Service Commissions (PSCs) is a forum formed with Chairman UPSC and Chairmen of 29 State PSCs to deliberate upon various issues concerning the Public Service Commissions in India.
 - The first National Conference of the Public Service Commissions was held in 1949. Thereafter the Conferences were held from time to time.
 - In 1999, Chairman UPSC became the ex-officio Chairman of the National Conference of Chairpersons of State Public Service Commissions. **Hence statement 2 is correct.**
 - These Conferences have striven to bring about some uniformity in the functioning of both the Union and the State PSCs. **Hence statement 1 is correct.**
 - Through the device of National Conference, a firm relationship has been attempted to be developed over the last 60 years among the State Public Service Commissions on matters concerning methods of recruitment, personnel policies, conduct of examinations etc.
 - Such gathering also provides a suitable platform for discussion on the changing socio-economic milieu and consequential changes to be brought about in the work ethos of the Commissions in consonance with expectations of the people and the constitutional obligations.

- Q 36. C**
- The Javadi Hills are an extension of the Eastern Ghats, located in the northern part of Tamil Nadu. They are known for their sandalwood and fruit trees and are a prominent feature of the Southern Indian landscape. **Hence, pair 1 is correctly matched.**
 - The Rajmahal Hills are located in the Santhal Pargana division of Jharkhand. They represent the northeastern-most part of the Indian Peninsular Plateau and are geographically considered part of

Eastern India. Hence, pair 2 is correctly matched.

- The Patkai Hills (or Patkai Bum) are located along the India-Myanmar border in the Northeast. They are part of the Purvanchal Range and span across states like Arunachal Pradesh and Nagaland. They are not in Western India. Hence, pair 3 is not correctly matched.
- The Dhauladhar range is part of the Lesser Himalayas and is located in the northern state of Himachal Pradesh. While Himachal Pradesh is in the western part of the Himalayan chain, the region is officially classified as Northern India. Hence, pair 4 is correctly matched.



- Q 37. D • **NarrowBand-Internet of Things (NB-IoT)** is a standards-based **low power wide area (LPWA) technology** developed to enable a wide range of new IoT devices and services. It is a wireless communication standard for the Internet of Things (IoT). NB-IoT belongs to the category of low-power wide-area networks (LPWAN), enabling to connect devices that need small amounts of data, low bandwidth, and long battery life. In NB-IoT applications, typically small data volume amounts need to be infrequently transmitted. Hence, statement 1 is correct.

- NB-IoT significantly **improves the power consumption of user devices**, system capacity and spectrum efficiency, especially in deep coverage. Battery life of more than 10 years can be supported for a wide range of use cases. **Hence, statement 2 is correct.**
- New physical layer signals and channels are designed to meet the demanding requirement of extended coverage – rural and deep indoors – and ultra-low device complexity. Initial cost of the NB-IoT modules is expected to be comparable to GSM/GPRS. The underlying technology is however much simpler than today's GSM/GPRS and its cost is expected to decrease rapidly as demand increases.
- Supported by all major mobile equipment, chipset and module manufacturers, NB-IoT can co-exist with 2G, 3G, and 4G mobile networks. It also benefits from all the security and privacy features of mobile networks, such as support for user identity confidentiality, entity authentication, confidentiality, data integrity, and mobile equipment identification.
- Recently BSNL, in partnership with Skylo, **has introduced worlds' first, satellite-based narrowband-IoT network in India.** This new 'Made in India' Solution, which is indigenously developed by Skylo, will connect with BSNL's satellite- ground infrastructure and provide PAN-India coverage, including Indian seas. The coverage will be so vast that it will not leave any dark patch within the boundary of India, from Kashmir & Ladakh to Kanyakumari, and from Gujarat to the North East, including the Indian seas. **Hence statement 3 is correct.**

- Q 38. B**
- **TReDS is an electronic platform for facilitating the financing / discounting of trade receivables of Micro, Small and Medium Enterprises (MSMEs)** through multiple financiers. These receivables can be due from corporates and other buyers, including Government Departments and Public Sector Undertakings (PSUs). **TReDS is a payment system authorised under the PSS Act.**
 - **In simple terms, a system like TReDS helps MSMEs get quick cash by selling their accepted invoices to banks, while also keeping the entire process transparent and safe.**
 - **Hence, statements 1 and 2 are correct.**
 - It is a platform for **uploading, accepting, discounting, trading and settling invoices / bills of MSMEs** and facilitating both receivables as well as payables factoring (reverse factoring). MSME sellers, corporate and other buyers, including Government Departments and PSUs, and financiers (banks, NBFC-Factors and other financial institutions, as permitted) are direct participants in the TReDS and **all transactions processed under this system are "without recourse" to MSMEs.**
 - Sellers, buyers and financiers are the participants on a TReDS platform. **Only MSMEs can participate as sellers in TReDS.** Corporates, Government Departments, PSUs and any other entity can participate as buyers in TReDS. Banks, NBFC - Factors and other financial institutions as permitted by the Reserve Bank of India (RBI), can participate as financiers in TReDS. **Hence, statement 3 is not correct.**

- Q 39. D**
- The Sports Ministry has approved the inclusion of four Indigenous Games to be a part of Khelo India Youth Games. The games include: **Gatka, Kalaripayattu, Thang-Ta and Mallakhamba.**
Gatka: It originated from the state of Punjab. **Hence option 4 is correct.**
 - It is traditional fighting style of the Nihang Sikh Warriors.
 - It is synonymous with swords, wooden sticks, leather shields and other sharp edged weapons.
 - It is believed to have originated when sixth Sikh guru Hargobind Singh adopted 'Kirpan' for self defense during Mughal era and tenth Guru Gobind Singh made it compulsory for everyone to use

the weapons for self defense.

- Since 2008, after the formation of the Gatka Federation of India (GFI), proper rules for the game were formulated and now it is considered a sport.

Kalaripayattu: It is an Indian martial art and fighting style that originated in modern-day Kerala. **Hence option 3 is correct.**

- It came into prominence when the local warriors found their weapons ineffective against the Portuguese ones.
- The British promulgated a ban on Kalaripayattu in 1804 to suppress the guerilla model revolt of Pazhassi Kerala Varma.
- Some gurus practised this art in the interior villages of Malabar and those who learned from them such as Kottakkal Kanaran Gurukkal preserved it for posterity.

Thang-Ta (sword and spear): It originated from the state of Manipur. **Hence option 1 is correct.**

- It is Manipuri Martial Art which is dedicated to fighting skill and worship.
- The martial art (sword fighting) form is used both as self defence as well as sports.

Mallakhamba: The word Mallakhamb is derived from the words 'Malla' and 'Khamb' which means 'Gymnast' and 'Pole'.

- Originating in the 12th century, it is a traditional sport and requires the practitioner to perform gymnastic, aerial yoga postures and wrestling grips on standing or hanging wooden pole or cane.
- Mallakhamba participants perform gravity defying acrobatics atop a long pole or hanging ropes.
- It has been well-known across India and Madhya Pradesh and Maharashtra have been the hotspots of this sport. **Hence option 2 is correct.**

Q 40. B • The IUCN Red List of Ecosystems Categories and Criteria is a global standard for how they assess the status of ecosystems, applicable at local, national, regional and global levels. Assessments determine whether an ecosystem is not facing imminent risk of collapse, or whether it is vulnerable, endangered, or critically endangered.

Q 41. B • Externalities are those benefits or harms accruing to another person, firm or any other entity which occur because some person, firm or any other entity may be involved in an economic activity. If someone is causing benefits or good externality to another, the latter does not pay the former. For example, development of roadways and railways generate additional income opportunities for the people living along such areas. They are called positive externalities. If someone is inflicting harm or bad externality to another, the former does not compensate the latter. For example, pollution caused by the power plants. They are called negative externalities.

- **Hence option (b) is the correct answer.**

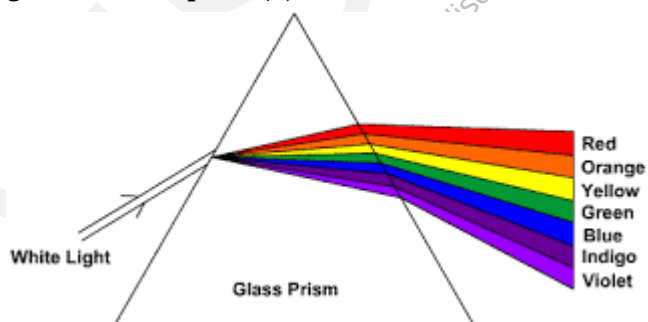
Q 42. C • Ground-level ozone is a secondary pollutant which is formed in the ground level through atmospheric reactions (Troposphere) in the presence of sunlight and responsible factors are high temperature and emissions of Oxides of Nitrogen (NOx) & Volatile Organic Compounds (VOCs). Major sources for NOx include vehicles, power plants and burning of fuel/waste and VOCs are emitted from vehicles, petrol pumps, use of solvents and burning of waste. **Hence, statement 1 is correct.**

- The ambient air quality standards for Ozone (O₃) has been notified by Central Pollution Control Board (CPCB) among one of the twelve (12) pollutants under National Ambient Air Quality Standard (NAAQS), 2009. The ambient air quality standards for Ozone (O₃) is prescribed as 100 µg/m³ for 8-hourly monitored value and 180 µg/m³ for 1-hourly monitored value for industrial, residential, rural and ecological sensitive area. **Hence, statement 2 is correct.**
- Sunlight (specifically UV light) is essential for the photochemical reactions that create ground-level ozone from its precursor pollutants. This is why ozone pollution is typically most problematic on hot, sunny days. **Hence, statement 3 is correct.**

Q 43. A The **correct chronological order for the Environmental Impact Assessment (EIA)** process in India is as follows:

- **Screening of the project:** This is the initial step to determine if a project requires an EIA and the level of assessment needed.
- **Preparation of EIA report and Environmental Management Plan (EMP):** The project proponent prepares a detailed report assessing potential environmental impacts and outlining mitigation measures.
- **Public consultation:** The public is given the opportunity to provide input and feedback on the proposed project and the draft EIA report.
- **Appraisal by the Expert Appraisal Committee (EAC):** The EAC evaluates the EIA report and the public consultation feedback to make a recommendation.
- **Grant of Environmental Clearance (EC):** Based on the EAC's recommendation, the regulatory authority decides whether to grant environmental clearance.
- **Hence, option (a) is the correct answer.**

Q 44. D • **Dispersion of Light:** This is the phenomenon where white light (like sunlight) splits into its constituent seven colors (VIBGYOR) when it passes through a transparent medium like a glass prism. Each color in white light has a different wavelength and travels at a different speed through glass. **Hence, option (d) is the correct answer.**



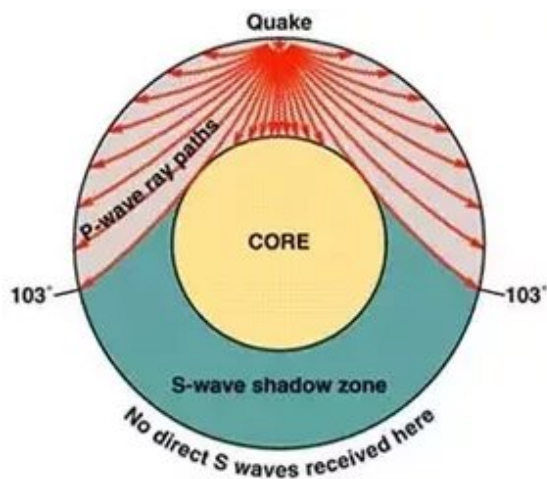
- **Other Options:**
- **Reflection:** This is the bouncing back of light from a surface (like a mirror).
- **Scattering:** This is the redirection of light in various directions by particles in the atmosphere (e.g., why the sky is blue).
- **Diffraction:** This is the slight bending of light as it passes around the edge of an object or through a narrow opening.

Q 45. C **Treaty of Tordesillas (1494)**

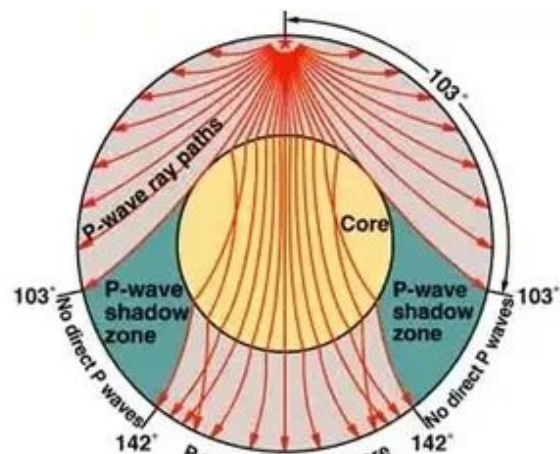
- In 1497, under the Treaty of Tordesillas (1494), **the rulers of Portugal and Spain** divided the non-Christian world between them by an imaginary line in the Atlantic, some 1,300 miles west of the Cape Verde Islands. **Hence option (c) is the correct answer.**
- Under the treaty, **Portugal could claim and occupy everything to the east of the line while Spain could claim everything to the west.** The situation was thus prepared for the Portuguese incursions into the waters around India.

- Q 46. A** • Community Reserves are established under the Wildlife (Protection) Amendment Act, 2002, on private or community land where the local people have given their consent for conservation efforts. **Hence, statement 1 is correct.**
- Conservation Reserves are also notified under the Wildlife (Protection) Act, 1972, and are specifically intended to act as buffer zones, corridors, or connectivity areas between existing national parks and wildlife sanctuaries. **Hence, statement 2 is correct.**
 - Biosphere Reserves (BRs) in India are not legally notified under the Wildlife (Protection) Act, 1972. They are designated by the Central Government, often following UNESCO's Man and the Biosphere (MAB) Programme guidelines. While they contain protected areas, the BR as a whole is not a statutory category under that Act. **Hence, statement 3 is not correct.**
- Q 47. C** • The edge effect is a well-documented ecological phenomenon where the boundary zone (ecotone) between two different adjacent ecosystems often exhibits greater species diversity and biological activity than the interior of either ecosystem. This is because it contains species from both bordering communities, as well as unique species adapted to the edge conditions. **Hence, statement 1 is correct.**
- Edge habitats typically favour generalist species that can thrive in a variety of conditions and are often negatively impacted by specialist interior species, which are sensitive to changes in light, temperature, and predation risk that are common in edge environments.
 - Habitat fragmentation, which breaks large continuous habitats into smaller, isolated patches, significantly increases the amount of edge habitat relative to the interior habitat in a landscape. This leads to a greater overall impact of edge effects across the ecosystem. **Hence, statement 2 is correct.**
- Q 48. C** • Apart from meeting the general requirements, Indian Remote Sensing missions are based on specific thematic applications like natural resources monitoring, ocean and atmospheric studies and cartographic applications. They are used for (i) Land/water resources applications (RESOURCESAT series and RISAT series); (ii) Ocean/atmospheric studies (OCEANSAT series, INSAT-VHRR, INSAT-3D, Megha-Tropiques and SARAL); and (iii) Large scale mapping applications (CARTOSAT series). **Hence statement 1 is correct.**
- The Indian National Satellite (INSAT) system is one of the largest domestic communication satellite systems in Asia-Pacific region with nine operational communication satellites placed in Geo-stationary orbit. The INSAT system with more than 200 transponders in the C, Extended C and Ku-bands provides services to telecommunications, television broadcasting, satellite newsgathering, societal applications, weather forecasting, disaster warning and Search and Rescue operations. **Hence statement 2 is correct.**

- Q 49. D** • **Trophic cascades are powerful indirect interactions** that can control entire ecosystems, occurring when a trophic level in a food web is suppressed. For example, a **top-down cascade will occur if predators are effective enough in predation to reduce the abundance**, or alter the behavior of their prey, thereby releasing the next lower trophic level from predation (or herbivory if the intermediate trophic level is a herbivore).
- A **top-down cascade is a trophic cascade where the top consumer/predator controls the primary consumer population**. In turn, the primary producer population thrives. The removal of the top predator can alter the food web dynamics. In this case, the primary consumers would overpopulate and exploit the primary producers. Eventually there would not be enough primary producers to sustain the consumer population. **Hence statement 2 is correct.**
 - These trophic interactions shape patterns of biodiversity globally. Humans and climate change have affected these cascades drastically. One example can be seen with sea otters on the Pacific coast of the United States of America. Over time, **human interactions caused a removal of sea otters**. One of their main prey, the pacific purple sea urchin eventually began to overpopulate. The overpopulation caused increased predation of giant kelp. As a result, there was **extreme deterioration of the kelp forests along the California coast**. **Hence statement 1 is not correct.**
- Q 50. A** • The Bird Atlas (notably the State-level initiatives like the Kerala Bird Atlas) is a massive citizen-science project. It relies on a network of volunteer birdwatchers to systematically document the distribution and abundance of birds across various habitats. **Hence, statement 1 is correct.**
- The primary utility of these atlases is to create a robust baseline for identifying biodiversity hotspots, understanding species richness, and establishing priority areas for conservation and future research. **Hence, statement 2 is correct.**
 - The atlas projects cover all species found in the region, including both resident and migratory birds. For example, the Kerala Bird Atlas documented 361 species, ranging from very rare to most abundant, and actually found higher species counts during the dry season (when many residents are active) than the wet season. **Hence, statement 3 is not correct.**
- Q 51. C** • Earthquake waves (seismic waves) are essentially of two types: Body waves (which travel through the interior of the earth) and Surface waves. Body waves are further divided into P-waves and S-waves, and their behavior tells us a lot about the Earth's internal structure.
- P-waves (Primary waves) are longitudinal waves, similar to sound waves. They move particles to and fro in the direction of propagation. Because they are compression waves, they can travel through all mediums: solids, liquids, and gases. They are the fastest seismic waves and the first to arrive at a seismograph station. **Hence, statement 1 is correct.**
 - S-waves (Secondary waves) are transverse waves, meaning they vibrate the ground up and down or side-to-side, perpendicular to the direction of the wave. A critical characteristic of S-waves is that they cannot travel through fluids (liquids or gases). Since the Earth's outer core is in a liquid state, S-waves are blocked by it. This creates a large "S-wave shadow zone" beyond 103° from the earthquake's epicenter. **Hence, statement 2 is correct.**



S-wave shadow zone = 154°



P-wave shadow zone = $103^\circ - 142^\circ$

- Q 52. A • Synthetic textiles like polyester and nylon are a significant source of microplastic pollution. During washing, tiny plastic fibres shed from these garments and enter waterways, as they are not biodegradable. **Hence, statement 1 is correct.**
- Textile dyeing and processing are major sources of water pollution. The effluents often contain harmful chemicals, heavy metals (like chromium and copper), and toxic dyes, which pose serious environmental and health risks. **Hence, statement 2 is correct.**
 - While natural fibres are generally more eco-friendly than synthetics, their production and processing are not entirely free of environmental impact. Cotton farming, for instance, often requires large amounts of water and pesticides, and processing natural fibres can also involve chemical treatments. **Hence, statement 3 is not correct.**
- Q 53. A • **Recently, the UN General Assembly (UNGA) announced the establishment of two mechanisms within the UN to promote international cooperation on the governance of Artificial Intelligence (AI). These are the 'Independent International Scientific Panel on Artificial Intelligence' and the 'Global Dialogue on AI Governance'. Hence, option (a) is the correct answer.**
- It builds on the "Pact for the Future" and the "Global Digital Compact," outlining a significant step towards international, non-military AI governance.
- Q 54. A • The 12th and the 13th centuries saw the emergence of the Kakatiyas. They were at first the feudatories of the Western Chalukyas of Kalyana, ruling over a small territory near Warangal. The founders of Vijayanagara Empire Harihara and Bukka were feudatories of Kakatiyas. **Hence, option (a) is not correct.**
- The kingdom saw powerful kings like Ganapatideva, Rudradeva and Prataparudra as well as the first-ever woman ruler in the subcontinent Rudramadevi. The Kakatiyas ruled from Hanumakonda in the beginning and shifted their capital to Warangal later.
 - The Kakatiya state was a military-state ready to face the threat of internal and external enemies. The military organization of the Kakatiyas was based on Nayamkara system. In this system, the ruler assigned fiefs to the Nayaks in lieu of their salaries and the Nayaks were to maintain some army for the ruler's use.
 - The Kakatiya dynasty expressed itself best through religious art. The Kakatiya temples, dedicated mostly to Siva, reveal in their construction a happy blending of the styles of North India and South

India which influenced the political life of the Deccan. The most important of these temples are those at Palampeta, Hanamkonda and the incomplete one in the Warangal fort. The temple at Palampeta, described as the 'brightest gem in the galaxy of Medieval Deccan temple architecture', was constructed by Recherla Rudra, a general of Kakatiya Ganapati, in S.1135 (A.D.1213).

- The Kakatiyas are known for their irrigation public works, sculpture and fire arts. Thanks to the well-planned irrigation facilities and a perfect system of chain tanks to suit the undulating nature of the terrain, the Kakatiya kingdom flourished economically leading to cultural progress also.
- Kakatiyas recognized the importance of long-distance trade. One indication that they wanted to encourage maritime trade, comes from the famous Mompalli epigraph. Motupalli was the chief port of the Kakatiyas and this port was visited by the Venitian traveller, Marco Polo.

Q 55. A Features of the Charter Act of 1833

The act, first of all, threw open judicial positions to Indians and provided for the appointment of a law commission for the codification of laws.

- **Law commission was established in 1834 under the Chairmanship of Lord Macaulay which recommended codification of the Penal Code and the Criminal Procedure Code.**
- **The Governor-General of Bengal became the Governor-General of India with exclusive legislative powers.** The presidencies of Bombay and Madras were deprived of their legislative powers. The Governor-General of India was given civil and military powers.
- The Government of India was created for the first time having the authority over the entire territorial area possessed by the British in India. **The First Governor-General of India was Lord William Bentick.**
- The members of the Governor General's council were reduced by the Pitt's India act 1784 was again increased to 4. The fourth member had very limited powers, he was not entitled to act as a member of the council except for legislative purposes.
- The Governor General Council had the authority to amend, repeal or alter any law in the entire length and breadth of India for any British, Foreigner or Indian.
- The activities of East India Company as a commercial body came to an end. The company purely became an administrative body. The company's territories in India were to be held by it "in trust for his Majesty, his heirs and successors".
- The act attempted to introduce a system of open competition for selection in Civil Services. It stated that Indians should not be debarred from holding any place, office and employment under the company. It was nullified after opposition from the Court of Directors.
- **Hence option (a) is the correct answer.**

Q 56. A The ocean floor is not a flat plain but contains diverse relief features similar to those found on land. As we move from the shoreline toward the open ocean, the depth increases in a specific sequence:

Continental Shelf (Shallowest)

- This is the extended margin of each continent occupied by relatively shallow seas and gulfs. It is the shallowest part of the ocean, with an average depth of about 30–200 meters. It has a very gentle slope.

Continental Slope

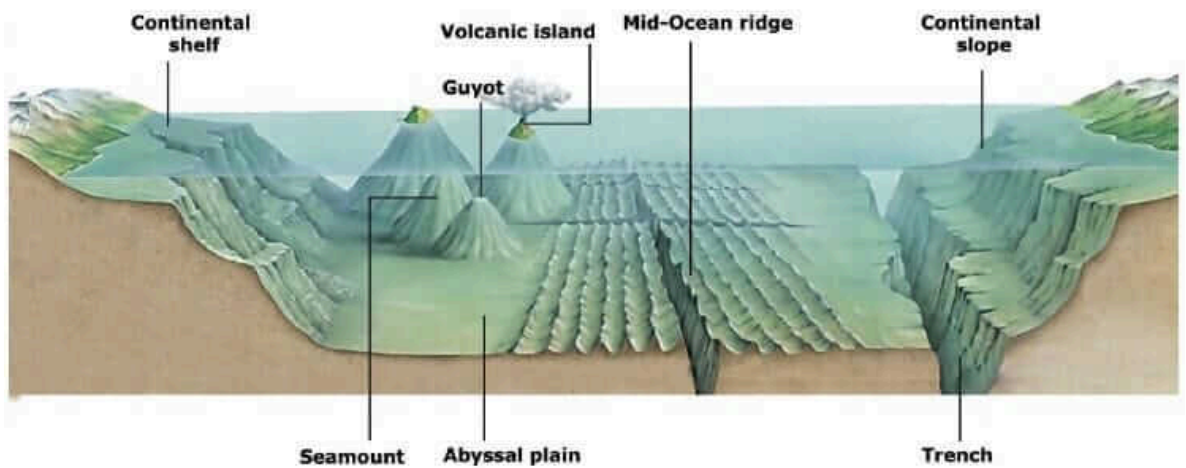
- The continental slope connects the continental shelf and the ocean basins. It begins where the bottom of the continental shelf sharply drops off. The gradient is much steeper here, and the depth typically ranges from 200 to 3,000 meters.

Abyssal Plain

- These are extensive, nearly level areas of the deep ocean floor. They are among the flattest and smoothest regions on Earth due to the thick accumulation of fine-grained sediments. Their depth usually ranges between 3,000 and 6,000 meters.

Oceanic Trench (Deepest)

- These are the deepest parts of the oceans. They are relatively steep-sided, narrow basins that occur where tectonic plates subduct. They can reach depths of more than 10,000 meters (e.g., the Mariana Trench in the Pacific Ocean is about 11,000 meters deep).



- Hence, option (a) is the correct answer.

Q 57. B • Recently, an amendment to the Customs Tariff Act (1975) was introduced to provide an HS (Harmonised System) code for GI-recognised rice varieties.

- The Harmonized System (HS) is an international standard for classifying traded products using numerical codes, developed by the World Customs Organization (WCO) to simplify customs, taxation, and trade statistics globally, allowing countries to uniformly identify goods for duties, tariffs, and regulations.
- HS is a global product classification system developed by World Customs Organization (WCO). It assigns specific 6-digit codes for varying classifications and commodities. **Hence, option (b) is the correct answer.**

Q 58. B • **‘FDI’ means investment through equity instruments** by a person resident outside India in an unlisted Indian company; or in **10% or more of the post issue paid-up equity capital on a fully diluted basis of a listed Indian company.** In case an existing investment by a person resident outside India in equity instruments of a listed Indian company falls to a level below 10% of the post issue paid-up equity capital on a fully diluted basis, the investment shall continue to be treated as FDI. Foreign investment is regulated through the **Foreign Exchange Management Act, 1999** as amended from time to time (FEMA) and rules/regulations issued thereunder.

FDI is prohibited in:

- Lottery Business including Government/private lottery, online lotteries, etc.

- Gambling and Betting including casinos etc.
- **Chit funds**
- **Nidhi company**
- **Trading in Transferable Development Rights (TDRs)**

Real Estate Business or Construction of Farm Houses

- ‘Real estate business’ shall not include development of townships, construction of residential /commercial premises, roads or bridges and Real Estate Investment Trusts (REITs) registered and regulated under the SEBI (REITs) Regulations 2014. **Foreign Direct Investment (FDI) in Real Estate Investment Trusts (REITs) in India is permitted up to 100%** under the automatic route, allowing foreign investors to invest in income-generating real estate.
- Manufacturing of cigars, cheroots, cigarillos and cigarettes, of tobacco or of tobacco substitutes
- Activities/sectors not open to private sector investment e.g. Atomic Energy and Railway operations (other than permitted activities mentioned in permitted sectors).
- Foreign technology collaboration in any form including licensing for franchise, trademark, brand name, management contract is also **prohibited for Lottery Business, Gambling and Betting activities.**
- **Hence, option (b) is the correct answer.**

- Q 59. B**
- Abhinaya Darpana is a foundational Sanskrit treatise on Indian classical dance and drama, attributed to the ancient scholar **Nandikeshvara around 1000 AD**. Often translated as "The Mirror of Gesture," it serves as a practical guide for performers, focusing on expressive techniques known as abhinaya
 - The Nāṭya Shāstra is a Sanskrit treatise on the performing arts. The text is attributed to sage Bharata, and its first complete compilation is dated to between 200 BCE and 200 CE, but estimates vary between 500 BCE and 500 CE. This is a foundational Sanskrit treatise on the performing arts, including drama, dance, and music, traditionally attributed to the sage Bharata Muni. **Hence, Pair 1 is not correctly matched.**
 - Abhinavabharati is a Sanskrit commentary on the Natyasastra, an ancient treatise on dramatic theory by Bharata Muni. It is the oldest extant commentary on the treatise. The Abhinavabharati was written by Abhinavagupta, the great Kashmiri Saivite spiritual leader and a yogi. **Hence, Pair 2 is correctly matched.**
 - The Gita Govinda is a famous 12th-century Sanskrit poem by the poet Jayadeva that lyricizes the divine love, separation, and ultimate reunion of Radha and Krishna. Structured into twelve chapters (cantos) containing 24 songs (ashtapadis), it is a foundational text of Vaishnava devotion, often interpreted as an allegory for the human soul's yearning for the divine. **Hence, Pair 3 is not correctly matched.**

- Q 60. B**
- Marketable securities are highly liquid financial instruments which can be easily traded on public exchanges, characterized by high volatility, readily available market prices, and quick conversion to cash.
 - Non-marketable securities cannot be sold on secondary markets, offering low liquidity, stable pricing based on intrinsic value, and typically requiring redemption at maturity.

- **The Central Government Debt** includes total outstanding liabilities of the Central Government on the security of the Consolidated Fund of India, including external debt valued at current exchange rates (Public Debt), total outstanding liabilities in the Public Account of India (Other Liabilities) and liabilities on account of Extra Budgetary Resources (EBR) raised by issuing Government of India (GoI) fully serviced bonds, adjusted for cash balance.
- **Public debt** is further classified into internal and external debt. Within internal debt, marketable debt comprises of Government dated securities and Treasury Bills, issued through auctions. **Hence option 1 is not correct.**
- **The non-marketable securities** in internal debt are the special **Central Government securities issued to NSSE, securities issued to international financial institutions**, special securities issued against securitisation of balances under postal insurance and annuity funds (POLIF and RPOLIF), compensation & other bonds (including **outstanding amounts under Sovereign Gold Bond Scheme** and Gold Monetisation Scheme), **special securities issued to public sector banks / EXIM Bank / IDBI Bank / IIFCL** and 14- day Intermediate Treasury Bills.
- **Hence, options 2 and 3 are correct.**

Q 61. D • National Bank for Agriculture and Rural Development (NABARD) was established in 1982 by an Act of the Parliament. NABARD, as a Development Bank, is mandated for providing and regulating credit and other facilities for the promotion and development of agriculture, small scale industries, cottage and village industries, handicrafts and other rural crafts and other allied economic activities in rural areas.

Credit related functions of NABARD are as follows:

- **Refinance to Rural Financial Institutions for investment credit** (long term loan) and production and marketing credit (short term loan) purposes for farm and off-farm activities in rural areas.
- **Providing long-term refinance to eligible NBFCs (including NBFC-mFIs)** to supplement their resources for agricultural, allied, and rural development lending.
- **Loans to State Governments for developing rural infrastructure** and strengthening of the Cooperative Credit Structure
- Loans for warehousing infrastructure to State Governments, State/ Central government Owned/ assisted entities, Cooperatives, Federation of cooperatives, Farmers' Producers Organizations, (FPOs), Federations of Farmers' Collectives, Primary Agricultural Credit Societies (PACS) / Cooperative Marketing Societies (CMS) or similar institutions, Corporates/ Companies, Individual entrepreneurs, etc.
- **Direct lending to Cooperatives and Producers' Organization**, support to State owned institutions /corporations under NABARD Infrastructure Development Assistance

Non-credit related functions of NABARD are as follows:

- Assist in policy formulation of GoI, RBI and State Governments on matters related to agricultural credit and rural development.
- Statutory **inspection/supervision of Regional Rural Banks (RRBs), State Cooperative Banks** and District Central Cooperative Banks (DCCBs), voluntary inspection of State Cooperative Agriculture and Rural Development Banks (SCARDBs) and their off-site surveillance.
- Promotional and developmental initiatives in the areas of farm, off-farm, micro finance, financial inclusion, Convergence with Govt sponsored programmes.

- Hence, option (d) is the correct answer.

Q 62. C • Gel electrophoresis: Gel electrophoresis is a laboratory method **used to separate mixtures of DNA, RNA, or proteins** according to molecular size. **Hence statement 1 is correct.**

- In Gel electrophoresis, the **molecules to be separated are pushed by an electrical field** through a gel that contains small pores. The molecules travel through the pores in the gel at a speed that is inversely related to their lengths. This means that a small DNA molecule will travel a greater distance through the gel than will a larger DNA molecule. **Hence statement 2 is correct.**
- As previously mentioned, gel electrophoresis involves an electrical field; in particular, this field is applied such that one end of the gel has a positive charge and the other end has a negative charge.
- Because **DNA and RNA are negatively-charged molecules**, they will be pulled toward the positively charged end of the gel. **Proteins, however, are not negatively charged**; thus, when one wants to separate proteins using gel electrophoresis, they must first mix the proteins with a detergent called sodium dodecyl sulfate. This treatment makes the proteins unfold into a linear shape and coats them with a negative charge, which allows them to migrate toward the positive end of the gel and be separated.
- Finally, after the DNA, RNA, or protein molecules have been separated using gel electrophoresis, bands representing molecules of different sizes can be detected.

Q 63. B • Statement 1 is incorrect: The National Commission for Protection of Child Rights (NCPCR) was set up in March 2007 under the Commission for Protection of Child Rights Act, 2005.

- **Statement 2 is correct:** The Commission's Mandate is to ensure that all Laws, Policies, Programmes, and Administrative Mechanisms are in consonance with the Child Rights perspective as enshrined in the constitution of India and also the UN Convention on the Rights of the Child.
- **Statement 3 is correct:** The Commission takes up specific complaints that come up before it for redressal of grievances and also takes up suo moto cases, it summons the violators of child rights, gets them presented before the Commission and recommends to the Government or the Judiciary, and takes action based on an inquiry.

Q 64. A • In a bid to put an end to vehicle thefts, the government has decided to launch a new standard for automobile industry, which involves use of a new technology.

- In this technique, **thousands of small dots** laser etched will be sprayed with a vehicle identification number on all over the vehicle's body. This will also include their engines.
- **This technology is termed as MicroDots** and it is almost impossible to remove these dots.
- Most of the vehicles stolen go untraced since auto-lifters take out the engines and other valuable parts before destroying the vehicle.
- Since, using MicroDots, almost all parts of the body inscribe the vehicle identification number or chassis number, it becomes impossible to change the identity of the stolen vehicles.
- Hence, the chances of vehicle recovery increase manifold using this technology. **Hence option (a) is correct.**

Q 65. B • United States (North), Mexico (West and South), Cuba (South-east) share the boudaries with the Gulf of Mexico. Hence, option (b) is the correct answer.



- Q 66. B** The Nilgiri tahr, lion-tailed macaque, and Andaman wild pig are all species endemic to India.
- The Nilgiri tahr is an endangered caprine species. This mountain ungulate is endemic to the Western Ghats of southern India, specifically the states of Tamil Nadu and Kerala. **Hence, point 1 is correct.**
 - The lion-tailed macaque is an endangered Old World monkey. This primate is endemic to the tropical evergreen rainforests of the Western Ghats in Karnataka, Kerala, and Tamil Nadu. **Hence, point 2 is correct.**
 - This is a subspecies of wild pig endemic exclusively to the Andaman Islands of India. **Hence, point 3 is correct.**
 - The snow leopard is found across a much wider range in Central and South Asia, including countries like Afghanistan, Bhutan, China, Kazakhstan, Kyrgyzstan, Mongolia, Nepal, Pakistan, Russia, Tajikistan, and Uzbekistan, not just India. **Hence, point 4 is not correct.**
- **Hence, option (b) is the correct answer.**

- Q 67. D**
- In the beginning, Mahavira led the life of a householder, but in the search for truth, he abandoned the World at the age of 30 and became an ascetic. He would not stay for more than a day in a village and for more than five days in a town. During the next twelve years, he meditated, practiced austerities of various kinds, and endured many hardships. In the thirteenth year, when he had reached the age of 42, he attained kaivalya (Jnan). Through **kaivalya** he conquered misery and happiness. Because of this conquest, he is known as Mahavira or the great hero or Jina. i.e. the conqueror and his followers are known as Jainas. **Kaivalya meaning enlightenment, not nirvana. Kaivalya could lead to nirvana.**
 - Jainism **recognized the existence** of the gods but **placed them lower than the Jina.**
 - **It did not condemn the varna system**, as Buddhism did. According to Mahavira, a person is born in a high or in lower varna in consequence of the sins or the virtues acquired by him in the previous birth.

- Mahavira looks for human values even in a chandala. In his opinion, through pure and meritorious life members of the lower castes can attain liberation.
- Jainism mainly aims at the attainment of freedom from worldly bonds.
- No ritual is required for acquiring such liberation.

It can be obtained through the right knowledge, right faith, and right action.

- These three are considered to be the **Three Jewels or Triratna of Jainism**.
- Jainism **prohibited the practice of war** and even **agriculture** for its followers because both involve the killing of living beings. Eventually, the Jainas mainly confined themselves to trade and mercantile activities.
- **Hence option (d) is the correct answer.**

Q 68. A • Phloem is the living vascular tissue responsible for the translocation of organic nutrients (mainly sucrose) throughout a plant.

- **Pressure Flow (Mass Flow) Hypothesis: This is the most widely accepted theory for phloem transport. It suggests that sugar is actively loaded into the phloem at the "source" (leaves), which creates a high concentration of solutes. This causes water to enter the phloem by osmosis, creating high turgor pressure that "pushes" the sap toward the "sink" (roots or fruits) where pressure is lower. Hence, option (a) is the correct answer.**
- **Energy Requirement:** Phloem transport is an active process. While the physical flow of sap is driven by pressure, the process of "loading" sucrose into the sieve tubes at the source and "unloading" it at the sink requires ATP (metabolic energy).
- **Direction of Flow:** Unlike xylem, which moves water strictly upward (unidirectional), phloem transport is bidirectional. Nutrients move from sources to sinks, which can be downward (to roots) or upward (to developing flowers, fruits, or new leaves).
- **Location and Extent:** Phloem is not limited to the leaves; it forms a continuous vascular network throughout the entire plant, including the stem, branches, and roots, to ensure every cell receives nutrients.

Q 69. C • **Censure Motion:** It is also a form of censure against the government, for **censuring the council of ministers** for specific policies and actions. It should **state the reasons** for its adoption in the Lok Sabha and **can be moved against an individual minister or a group of ministers or the entire council of ministers**.

A no-confidence motion is a parliamentary procedure whereby the legislature expresses its lack of confidence in the government, thereby forcing it to resign or face dissolution. This means that the government can be removed from power if the Lok Sabha passes a motion of no confidence. **It is passed against the entire Council of Ministers only.**

- A motion of no confidence is a formal statement by the Lok Sabha that it no longer has confidence in the government. If the motion is passed, the President of India must dissolve the Lok Sabha and call for new elections. **There is no obligation to state specific reasons for moving the motion.**
- The **Indian Constitution does not explicitly mention or define a no-confidence motion. However, the concept is implied through Article 75(3), which states that the Council of Ministers shall be collectively responsible to the Lok Sabha.**

- If the Speaker decides to admit the no-confidence motion, it will be debated in the Lok Sabha. If the motion is passed by a majority of the members present and voting, the government will be dismissed.

Out of all NCMs, only three have resulted in the actual fall of the government. These are the following:

- The V P Singh government in 1990
- The H D Deve Gowda government in 1997
- The Atal Bihari Vajpayee government in 1999

Q 70. C Every metropolitan area shall have a metropolitan planning committee to prepare a draft development plan for the Metropolitan area. **Hence statement 1 is correct.** The state legislature may make provisions with respect to the following:

- The composition of such committees;
- The manner of election of members of such committees;
- The representation in such committees of the Central government, state government and other organisations;
- The functions of such committees in relation to planning and coordination for the metropolitan area; and
- The manner of election of chairpersons of such committees.
- The Act lays down that **two-thirds of the members of a metropolitan planning committee** should be elected by the elected members of the municipalities and chairpersons of the panchayats in the metropolitan area from amongst themselves. The representation of these members in the committee should be in proportion to the ratio between the population of the municipalities and the panchayats in that metropolitan area. **Hence statement 2 is correct.**

Q 71. B • Edible oils form an essential component of India's food and nutritional security, and oilseeds play a vital role in the livelihoods of millions of farmers. India's total edible oil production was recorded at 12.18 million tonnes during 2023-24 (latest data). The country is able to meet only 44 percent of its domestic demand for edible oils through internal production. **Despite being among the world's largest producers of oilseeds, India remains substantially dependent on imports to bridge its edible oil deficit.** Notably, **import dependence on edible oils** has declined from 63.2% in 2015-16 to **56.25% in 2023-24**, reflecting a modest improvement in self-sufficiency from 36.8% to 43.74%. **Hence, statement 2 is correct.**

- Domestically, **oilseeds hold the second-highest acreage and production value in Indian agriculture after food grains.** Nine major oilseeds, groundnut, soybean, rapeseed-mustard, sunflower, sesame, safflower, niger, castor, and linseed, occupy 14.3% of the gross cropped area, contribute 12–13% of dietary energy, and account for about 8% of agricultural exports. **Hence, statement 1 is not correct.**
- **Oil Palm has highest vegetable oil yielding capability per hectare.** It produces two distinct oils, i.e. palm oil and palm kernel oil, which are used for culinary as well as industrial purposes. In comparative terms, the yield of palm oil is 5 times the yield of edible oil obtainable from traditional oilseeds. **Andhra Pradesh and Telangana are major Oil palm growing States and account 98% of total production.**

- Nine major oilseeds account for 14.3% of the yearly gross cropped area in India, contribute 12-13% to the dietary energy and account for about 8% of the agricultural exports. **India ranks first in the production of castor, safflower, sesame and niger, second in groundnut, third in rapeseed-mustard, fourth in linseed, and fifth in soybean.** Rajasthan, Madhya Pradesh, Gujarat, and Maharashtra are the major oilseed-producing states, contributing to more than 77% of the total oilseed production in the country. **Hence, statement 3 is correct.**

Q 72. B • The 9th Round of India–Peru Trade Agreement negotiations was successfully concluded in Peru.

Geographical features:

- Location: Western South America
- Capital: Lima
- **Borders: North:- Ecuador, Colombia; East: Brazil; Southeast:- Bolivia; South: Chile**
- Bordering water body (West): Pacific Ocean
- Highest Peak: Mount Huascarán (6,768 m)
- Lake Titicaca: Shared with Bolivia; world's highest navigable lake (12,500 ft).

Economy (as per Confederation of Indian Industry):

- Peru has emerged as India's 3rd largest trading partner in the Latin American and Caribbean region
- India is a key importer of Peruvian gold and copper.
- Peru is also actively engaged in regional blocs, being an observer of SICA, an associate member of Mercosur, and a full member of the Pacific Alliance.

Q 73. A • Of the five states, the matter can be in, the Bose-Einstein condensate is perhaps the most mysterious. Gases, liquids, solids, and plasmas were all well studied for decades, if not centuries; Bose-Einstein condensates weren't created in the laboratory until the 1990s.

- **A Bose-Einstein condensate is a group of atoms cooled to within a hair of absolute zero.** When they reach that temperature the atoms are hardly moving relative to each other; they have almost no free energy to do so. At that point, the atoms begin to clump together and enter the same energy states. They become identical, from a physical point of view, and the whole group starts behaving as though it were a single atom.
- This state was first predicted, generally, in 1924–1925 by **Albert Einstein** following and crediting a pioneering paper by **Satyendra Nath Bose** on the new field now known as quantum statistics.
- **One application for BEC is for the building of so-called atom lasers, which could have applications ranging from atomic-scale lithography to measurement and detection of gravitational fields. Hence the correct option is (a)**

Q 74. D • An eclipse happens when a planet or a moon gets in the way of the Sun's light. A happens when the Moon gets in the way of the Sun's light and casts its shadow on Earth.

Solar Eclipse:

- When the sun, moon, and earth come in a straight line and the moon is between the sun and the earth, the shadow of the moon falls on the earth. The sun is not fully visible to the people of the part of the Earth where the moon is cast. Then there is a solar eclipse for them. **Whenever a**

solar eclipse occurs, it happens on the day of the new moon, because the moon comes between the Earth and the sun on the new moon. **Hence, statement 1 is not correct.**

- But not everyone experiences every solar eclipse. Getting a chance to see a total solar eclipse is rare. The Moon's shadow on Earth isn't very big, so only a small portion of places on Earth will see it. You have to be on the sunny side of the planet when it happens. You also have to be in the path of the Moon's shadow. On average, **the same spot on Earth only gets to see a solar eclipse for a few minutes about every 375 years. Hence, statement 3 is not correct.**

Lunar Eclipse:

- When the sun, earth, and the moon come in a straight line and the earth is between the sun and the moon, the shadow of the earth falls on the moon, the part of the moon that shines with sunlight loses its luster, without the rays of the sun. This is called a lunar eclipse. Whenever **a lunar eclipse occurs, it happens only on the full moon day because on this day, the Earth comes between the sun and the moon. Hence, statement 2 is not correct.**

Q 75. B • Nanomicelles are ultramicroscopic (size 10 – 100 nm) structures used in the treatment of disease. They have a hydrophilic outer shell and a hydrophobic exterior, making them capable of delivering poorly water-soluble drugs and protecting drug molecules.

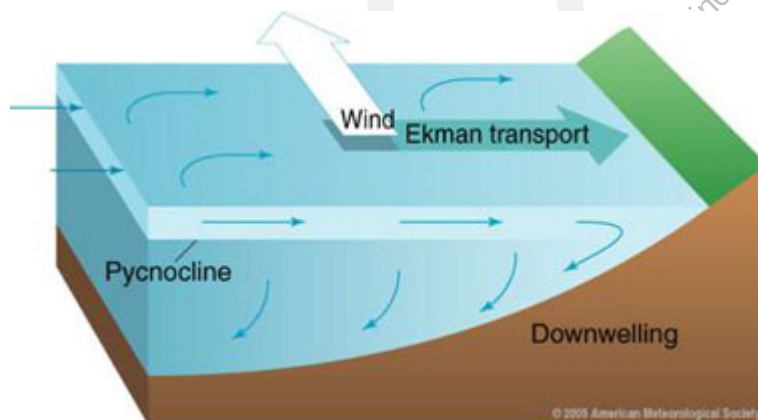
- Intensive research has shown that nano micelles could be used in a variety of medico-pharmacological instances to catalyze drug effects or improve treatment. Studies postulate that **nanomicelles are helpful in facilitating therapeutic solutions that involve protein and peptide delivery.**
- Bioengineers have taken great interest in the functionality of nanomicelles due to their **small size (10 – 100 nm), the capability to solubilize lipophilic drugs in various levels of drug loading, low toxicity, and ability to conjugate with target ligand and stimuli-sensitive regions.**
- However, because micelles disintegrate rapidly after tissue penetration, using nanomicelles as drug carriers become a challenge. Research suggests that in order to efficiently address this, prolonged blood circulation and control for the appropriate release of drugs on target sites should be modulated.
- Nanomicelles are also hypothesized to improve cancer treatment through overcoming drug resistance, improving anti-cancer drug efficacy, and reducing drug toxicity. **Hence the correct option is (b)**
- **Tardigrades** - often called water bears - are creatures under a millimeter long that can survive being heated to 150 Degree C and frozen to almost absolute zero. They were traveling on an Israeli spacecraft that crash-landed on the moon in April 2019.
- Recently, China live-streamed footage of its new manned submersible parked at the bottom of the Mariana Trench on Friday, part of a historic mission into the deepest underwater valley on the planet. The "**Fendouzhe**", or "**Striver**", descended more than 10,000 meters (about 33,000 feet) into the submarine trench in the western Pacific Ocean with three researchers on board.
- Also in a separate research, **Eurythenes plasticus, a newly described species of crustacean from the Marian Trench** in the Pacific Ocean, was named after pieces of plastic ingested by the animal.

Q 76. A • The regulating Act of 1773 was passed by the British Parliament to control the territories of the East India Company majorly in Bengal. This act was passed due to the misgovernment by the British

East India government that introduced a situation of bankruptcy and the government had to interfere with the affairs of the Company.

- The Regulating Act of 1773 vested the administration of British territories in India in the hands of a **Governor-General assisted by a Council of four members. Hence statement 1 is correct.**
- The Governor-General was to preside over meetings of the Council, **but the decision of the majority was to bind the whole, the Governor-General having merely a casting vote in case of an equal division. Hence statement 2 is not correct.**
- **Three members formed the quorum.**
- The Governor-General and the Councillors were named in the Act, **Warren Hastings was named as the first Governor-General and Clavering, Francis, Monson and Barwell as the four Councillors.**
- **At the time of his appointment Barwell was in India, already in the service of the Company. The other three Councilors reached India in October, 1774.**
- The new Councilors set out from England with a strong prejudice that Warren Hastings was corrupt and so was the entire government of the Company in Bengal. Of the Triumvirate Clavering, Francis and Monson - Francis was undoubtedly the ablest and the most ambitious. His ambition was to replace Warren Hastings or to succeed him.

- Q 77. C**
- Ekman transport describes the net movement of surface water at a 90-degree angle to the wind direction (to the right in the Northern Hemisphere and to the left in the Southern Hemisphere) due to the Coriolis effect. In subtropical regions, trade winds and westerlies drive water toward the center of the ocean basins. This creates a "hill" of water and high pressure, resulting in the large-scale circulating systems known as subtropical gyres. **Hence, statement 1 is correct.**



- La Niña is characterized by unusually strong trade winds that push warm surface water toward the Western Pacific (near Indonesia). **This allows cold, nutrient-rich deep water to rise to the surface along the coast of South America (Peru and Ecuador) in a process called upwelling.** Because this deep water is full of nutrients, it supports a massive increase in phytoplankton, which in turn leads to a fishing boom (particularly for anchovies). **Hence, statement 2 is correct.**

- Q 78. D**
- Algae is commercially cultivated for Pharmaceuticals, Nutraceuticals, Cosmetics and Aquaculture purpose. Humans use algae as food, for production of useful compounds, as biofilters to remove nutrients and other pollutants from wastewaters, to assay water quality, as indicators of environmental change, in space technology, and as laboratory research systems.
 - **Carbon Dioxide Fixation:** Like any other plant, algae, when grown using sunlight, consume (or absorb) carbon dioxide (CO₂) as they grow, releasing oxygen (O₂). For high productivity, algae

require more CO₂, which can be supplied by emissions sources such as power plants, ethanol facilities, and other sources.

- **Bio-fuel & Oil extraction:** Algae can be used to make Biodiesel (see algaculture), Bioethanol and biobutanol and can produce vastly superior amounts of vegetable oil, compared to terrestrial crops grown for the same purpose.
- **Purification of wastewater:** Algae thrive in nutrient-rich waters like municipal waste waters (sewage), animal wastes and some industrial effluents, at the same time purifying these wastes while producing a biomass suitable for biofuels production.
- **Food Supplements:** Algae are national foods of many nations: China consumes more than 70 species, including fat choy, a cyanobacterium considered a vegetable; Japan, over 20 species; Ireland, dulse; Chile, cochayuyo. Laver is used to make "laver bread" in Wales where it is known as bara lawr; in Korea, gim; in Japan, nori and aonori. It is also used along the west coast of North America from California to British Columbia, in Hawaii and by the Māori of New Zealand.

- Q 79. A** • The Hoysala temples, renowned for their intricate, soapstone, star-shaped architecture, are primarily located in Karnataka, India, and date from the 12th–13th centuries. The most prominent, UNESCO-recognised sites are the Channakeshava Temple in Belur, the Hoysaleswara Temple in Halebidu, and the Keshava Temple in Somanathapura.

Key temples associated with the Hoysala dynasty include:

- **Chennakesava Temple, Belur (1117 CE): Built by King Vishnuvardhana to commemorate his victory over the Cholas. Hence, option 1 is correct.**
 - **Hoysaleswara Temple, Halebidu (1121 CE): A massive twin-shrined temple dedicated to Lord Shiva.**
 - Keshava Temple, Somanathapura (1279 CE): A Trikuta (triple shrine) temple dedicated to Lord Krishna.
 - Amruteswara Temple, Amrithapura (1196 CE): Known for its exquisite, detailed carvings.
 - Lakshmi Devi Temple, Doddagaddavalli: An early, notable example of Hoysala style.
 - Buceswara Temple, Koravangala: Dedicated to Shiva.
 - **Mallikarjuna Temple, Basaralu: Known for its detailed, ornate architecture. Hence, option 4 is correct.**
 - Nageshwara-Chennakeshava Temple, Mosale: Features both Shiva and Vishnu shrines.
 - **Shore Temple: This temple is located in Mahabalipuram, Tamil Nadu. It was built by the Pallava dynasty in the 7th–8th centuries CE and is an iconic example of early Dravidian rock-cut architecture. Hence, option 2 is not correct.**
 - **Airavateswara Temple:** Located in Darasuram, this temple **was built by the Chola emperor Rajaraja II in the 12th century.** It is part of the "Great Living Chola Temples" UNESCO World Heritage Site. **Hence, option 3 is not correct.**
- **Hence, option (a) is the correct answer.**

- Q 80. C** • **ISRO LVM3-M5 Rocket Launch: ISRO is set to launch its largest rocket, LVM-3, to send a communication satellite, CMS-03, into space. This launch marks the first instance of ISRO deploying a satellite weighing over 4,000 kg from Indian soil into the geosynchronous transfer orbit (GTO).**

Details of CMS-03 Satellite

- The CMS-03 satellite weighs 4,410 kg.
- This launch showcases the growing capability of the LVM3 rocket.

LVM-3 Rocket Overview

- The LVM-3 rocket, formerly known as Geosynchronous Launch Vehicle Mark 3 or GSLV Mk 3, is versatile in its launch capabilities:
- Uses solid, liquid, and cryogenic-fuel based engines.
- Can carry up to 8,000 kg to low earth orbit (LEO) and up to 4,000 kg to geosynchronous orbit.

- Q 81. C**
- In terms of Section 20 of the **RBI Act 1934**, RBI has the **obligation to undertake the receipts and payments of the Central Government** and to carry out the exchange, remittance and other banking operations, including the management of the public debt of the Union. Further, as per Section 21 of the said Act, RBI has the **right to transact Government business of the Union in India**. State Government transactions are carried out by RBI in terms of the agreement entered into with the State Governments in terms of section 21 A of the Act. **Hence, statement 1 is correct.**
 - RBI carries out the general banking business of the governments through its own offices and commercial banks, both public and private, appointed as its agents. **Reserve Bank of India maintains the Principal Accounts of Central as well as State Governments at its Central Accounts Section, Nagpur.** A network comprising the Government Banking Divisions of RBI and branches of agency banks appointed under Section 45 of the RBI Act carry out the government transactions. **At present all the public sector banks** and select private sector banks act as RBI's agents. Only designated branches of agency banks can conduct government banking business. **Hence, statements 2 and 3 are correct.**
 - The Reserve Bank of India is the aggregator for accounting of all GST collections in the respective government accounts. Agency banks who collect the GST for challans generated by tax payers online on the GST portal report the collections for settlement to government accounts to RBI. RBI has also facilitated payment of GST by tax payers directly into government accounts at RBI by using **NEFT / RTGS payment options provided in GST portal.**
- Q 82. C**
- **Extended producer responsibility, a practice and a policy approach in which producers take responsibility for management of the disposal of products they produce once those products are designated as no longer useful by consumers.** Responsibility for disposal may be fiscal, physical, or a combination of the two.
 - Motivations for extended producer responsibility practices include a mixture of economic, environmental, and social factors. **Extended producer responsibility shifts the economic burden of the cost of disposal from the government to the producer of the product.** Within an environmental context, products must be designed for recyclability, and extended producer responsibility encourages design for recycling while discouraging the use of toxic components in the product. Finally, extended producer responsibility meets increasing consumer demand for environmentally friendly products that can easily be recycled or are manufactured using recycled content. Extended producer responsibility is a product-focused strategy that encourages environmentally friendly design and disposal of products through transfer of this responsibility to product producers.

- The intergovernmental Organisation for Economic Co-operation and Development's (OECD) definition of extended producer responsibility identifies two specific features: the shifting of responsibility for disposal "upstream" from municipalities to producers and encouragement through incentives to make the design of products more environmentally friendly.

- Q 83. C**
- **The premise of a bank's business model is staged around the concept of intermediation, which in other words is the process of connecting borrowers with lenders.** Banks charge a higher rate of interest on loans which they give out relative to the interest paid by them on deposits. By calculating the difference between these two rates of interest, one arrives at the Net Interest Income, which is the revenue generated by the bank through its lending activities.
 - **The CASA ratio** is a critical metric that represents the proportion of a bank's total deposits held in CASA accounts. This ratio indicates the bank's financial health and operational cost efficiency. **CASA deposits are typically low-cost funds for banks since they offer lower interest rates compared to fixed deposits.** A high CASA ratio reflects strong liquidity and stability, enabling banks to provide competitive products and services. **Hence, statement 1 is correct.**
 - **The Provision Coverage Ratio (PCR)** measures a bank's ability to cover its bad loans or non-performing assets (NPAs) with the money it has set aside for that very purpose. It is calculated by dividing the total value of the bank's provisions for bad loans by the total value of its non-performing assets. Provision Coverage Ratio is indicative of the bank's ability to absorb potential loan losses. **A higher ratio indicates that the bank has set aside more money to cover potential NPAs, and therefore has a stronger buffer against potential defaulters i.e., credit risk shocks. Hence, statement 2 is correct.**

- Q 84. C**
- Both plant and animal cells are eukaryotic, meaning they share several core organelles, but they also have distinct differences based on how they obtain energy and maintain their structure.
 - **Mitochondria (Common to Both):** Known as the "powerhouse of the cell," mitochondria are responsible for cellular respiration, where they convert glucose into energy (ATP). Both plants and animals need energy to perform cellular functions, so both possess mitochondria. **Hence, option (c) is the correct answer.**
 - **Cell Wall (Plant Only):** This is a rigid outer layer made of cellulose that provides structural support and protection. Animal cells lack a cell wall, which allows them to be more flexible and take on various shapes.
 - **Chloroplast (Plant Only):** These organelles contain chlorophyll and are the site of photosynthesis. Since animals are heterotrophs (they eat food) rather than autotrophs (they make food), they do not have chloroplasts.
 - **Plasmid (Primarily Prokaryotic):** Plasmids are small, circular DNA molecules typically found in bacteria. While they are occasionally found in some fungi or used in labs for genetic engineering in plants/animals, they are not a standard internal feature of typical plant or animal cells.

- Q 85. A**
- A volcano is a place where gases, ashes and/or molten rock material – lava – escape to the ground. A volcano is called an active volcano if the materials mentioned are being released or have been released out in the recent past.
- The layer below the solid crust is mantle. It has higher density than that of the crust. The mantle contains a weaker zone called asthenosphere. It is from this that the molten rock materials find

their way to the surface.

Volcanoes are classified on the basis of nature of eruption and the form developed at the surface. Major types of volcanoes are as follows:

Shield Volcanoes:

- Barring the basalt flows, the shield volcanoes are the largest of all the volcanoes on the earth. The Hawaiian volcanoes are the most famous examples. These volcanoes are mostly made up of basalt, a type of lava that is very fluid when erupted.
- For this reason, these volcanoes are not steep. They become explosive if somehow water gets into the vent; otherwise, they are characterised by low-explosivity. The upcoming lava moves in the form of a fountain and throws out the cone at the top of the vent and develops into cinder cone.

Composite Volcanoes:

- **These volcanoes are characterised by eruptions of cooler and more viscous lavas than basalt. Hence statement 1 is not correct.**
- These volcanoes often result in explosive eruptions. Along with lava, large quantities of pyroclastic material and ashes find their way to the ground. This material accumulates in the vicinity of the vent openings leading to formation of layers, and this makes the mounts appear as composite volcanoes.
- Composite volcanoes, also known as stratovolcanoes, are tall, steep-sided volcanoes formed from alternating layers of lava flows, ash, and other volcanic debris. **They are characterized by their distinctive conical shape and their explosive eruptions. Hence statement 2 is correct.**

Some of the features of composite volcanoes are:

- **Steep-sided cone shape:** Composite volcanoes have a tall, conical shape, with steep sides that can reach up to 30 degrees.
 - **Crater:** At the summit of the volcano, there is often a bowl-shaped depression called a crater. This is the vent through which lava, ash, and other volcanic materials are ejected during an eruption.
 - **Lava flow:** Composite volcanoes are typically composed of both explosive eruptions and effusive eruptions, which produce lava flows. **The lava flows from these volcanoes can be quite thick and viscous, meaning that they don't travel very far from the vent. Hence statement 3 is not correct.**
 - **Lahars:** Composite volcanoes are prone to producing lahars, which are fast-moving mudflows that are triggered by volcanic activity. Lahars can be highly destructive, as they can carry large boulders, trees, and other debris, and can travel many kilometers from the volcano.
- **Caldera:** These are the most explosive of the earth's volcanoes. They are usually so explosive that when they erupt they tend to collapse on themselves rather than building any tall structure. The collapsed depressions are called calderas. Their explosiveness indicates that the magma chamber supplying the lava is not only huge but is also in close vicinity.

Flood Basalt:

- **Provinces** These volcanoes outpour highly fluid lava that flows for long distances. Some parts of the world are covered by thousands of sq. km of thick basalt lava flows. There can be a

series of flows with some flows attaining thickness of more than 50 m. Individual flows may extend for hundreds of km.

- The Deccan Traps from India, presently covering most of the Maharashtra plateau, are a much larger flood basalt province. It is believed that initially the trap formations covered a much larger area than the present.
- **Mid-Ocean Ridge Volcanoes:** These volcanoes occur in the oceanic areas. There is a system of mid-ocean ridges more than 70,000 km long that stretches through all the ocean basins. The central portion of this ridge experiences frequent eruptions.

Q 86. C • Recent Context: In 2025, the United Nations Secretary General, as depository of the WHO Constitution, circulated a notification from the United States of America of its intention to withdraw from the World Health Organization, with a stated effective date of 22 January 2026. This notification is pending consideration by WHO's governing bodies.

- **UN Agency: The WHO is the United Nation's specialized health agency and is mandated to coordinate the world's response to global health threats.**
- **Genesis: The International Health Conference held in New York in 1946 adopted the Constitution of WHO, which entered into force in 1948. Hence, option (c) is the correct answer.**
- **Key Functions:** WHO coordinates the world's response to health emergencies, promote well-being, prevent disease and expand access to health care.

Q 87. C • Article 248 (1) of the Constitution of India says that the Parliament has exclusive power to make any law with respect to any matter not enumerated in List II and III. Such power shall include the power of making any law imposing a tax not mentioned in either of those lists. Residuary power of legislation includes the power to levy residuary taxes. As it is vested in Centre, Constitution has made the Centre very strong that shows the unitary character of Indian polity.

India has adopted some of the features from the Constitution of Canada like the

- Federation with a strong Centre
 - vesting of residuary powers in the Centre
 - the appointment of state governors by the Centre, and
 - advisory jurisdiction of the Supreme Court.
- **Hence both the statements are correct.**

Q 88. B Yakshagana

- Yakshagana is a traditional folk dance-drama performance that combines music, song, dance, scholarly dialogue, and colorful costumes. It is popular in the coastal districts of Karnataka and the Kasaragod district of Kerala.
- It is based on mythological stories and Puranas.
- Yakshagana has two main variations, each of which has many variations: Moodalapaya (the eastern form which is popular in north Karnataka) and Paduvalapaya (western style also known as coastal Yakshagana). Of the two, the coastal Yakshagana is more popular for the great sophistication that it has achieved over the years. It is more exuberant and refined when

compared to all the other styles. Paduvalapaya is performed in three coastal districts of Karnataka–Dakshina Kannada, Udupi and Uttara Kannada.

- In the region with Mangalore at its centre (which includes Mangalore, Sullia, Puttur, Sampaje, Bantwala, Belthangady, Karkala, Kasaragod (Kerala), Kodagu etc.) the tenkuthittu or the southern style is dominant. In Udupi district (spreading from Udupi to Kundapura and some parts of Uttara Kannada district), the badaguthittu or northern style holds sway. The extreme northern parts of Uttara Kannada district are known for the northern style (badaabadagu tittu).
- Themes are based on Mahabharata i.e. Draupadi swayamvar, Subhadra vivah, Abhimanyu vadh
- **Hence option (b) is the correct answer.**

Jatra:

- Jatra is a traditional theatre form from the Bengal region (West Bengal, Odisha, and Assam). It is known for its melodramatic and often lengthy performances that can run through the night. The themes of Jatra are typically derived from mythology, historical events, or social issues.
- Although Jatra includes music and drama, it does not follow the "Bhagavata" style, which specifically involves storytelling focused on the Bhagavata Purana and includes elements of classical dance. Jatra is more narrative-driven and less focused on religious or devotional content.

Mudiyettu

- It is a ritual dance drama from Kerala based on the mythological tale of a battle between the goddess Kali and the demon Darika. It is a community ritual in which the entire village participates. After the summer crops have been harvested, the villagers reach the temple in the early morning on an appointed day.

Tamasha:

- Tamasha is a traditional Marathi theatre form known for its lively, bawdy humor, and energetic performances. It often includes Lavani songs and dance, with themes ranging from romance to social commentary.
- Tamasha is a folk entertainment form that does not adhere to the "Bhagavata" style. It is more known for its entertaining, often satirical content, and is not focused on religious or devotional narratives like those in Yakshagana.

- Q 89. A** • The effective control of the Peshwas ended with the great defeat of Panipat (1761) at the hands of the Afghans and the death of the young Peshwa Madhav Rao I in 1772. **Thereafter the Marāṭha state was a confederacy of five chiefs under the nominal leadership of the Peshwa at Poona (Pune) in western India.** The other states were Sindhia, Holkar, Gaikwad, and Bhonsle. Though they united on occasion, as against the British (1775–82), more often they quarreled. The Confederacy expressed a general Marāṭha nationalist sentiment but was divided bitterly by the jealousies of its chiefs. **Hence pair 1 is not correctly matched.**
- **Bhonsle Dynasty:** They ruled from Nagpur in the present Maharashtra state and were a leading power in the 18th-century Marāṭhā confederacy. They were British clients from 1818 to 1853. Raghuji Bhonsle of Berār founded the dynasty in 1730. **Hence pair 5 is not correctly matched.**
 - **Gaikwad Dynasty:** Their capital was at Baroda in Gujarāt state. The state became a leading power in the 18th-century Marāṭhā confederacy. The founder of the dynasty was Damaji I who had risen to power by 1740. The last Gaekwad, Sayaji Rao III, died in 1939. **Hence pair 3 is not correctly matched.**

- **Holkar Dynasty:** The family, of peasant origin and of shepherd caste, was said to have migrated from the Mathurā district to the Deccan village of Hol, or Hal, the name of which, coupled with kar(“inhabitant of”), became the family surname. The dynasty's founder, Malhar Rao Holkar, rose from peasant origins by his own ability. In 1724 Bajji Rao I, the Peshwa (prime minister) of the Marathā state, gave him command of 500 horses, and he soon became the Peshwa's chief general in Malwa, with headquarters at Maheshwar and Indore. **Hence pair 4 is not correctly matched.**
- **Sindhia Dynasty:** The Maratha ruling family of Gwalior, which for a time in the 18th century dominated the politics of northern India. The dynasty was founded by Ranoji Sindhia, who in 1726 was put in charge of the Mālwa district by the Peshwa (prime minister of the Maratha state). At his death in 1750, Ranoji had established his capital at Ujjain; only later was the Sindhia capital moved to the rock fortress of Gwalior. **Hence pair 2 is correctly matched.**
- **Thus option (a) is the correct answer.**

- Q 90. D** • The Rabatak inscription is a crucial 2nd-century CE Bactrian-language text from Afghanistan, detailing the lineage and divine mandate of Kushan Emperor Kanishka I, revealing his claim to rule through deities like Nana, and marking the shift from Greek to "Aryan" language in official Kushan inscriptions, highlighting his power and conquests in Northern India.
- Discovered in 1993, this inscription provides invaluable historical insights, contradicting earlier assumptions about Kanishka and offering a political map of the Ganges Valley.
 - According to the Rabatak Bactrian inscription, the Kushana realm at the time of Kanishka extended to the cities of Saketa, Kausambi, Pataliputra, and Sri-Campa in the Ganges-Yamuna valley. A colossal statue of Kanishka near Mathura with a Brahmi inscription labeling him 'Great King, King of Kings, Son of God, Kanishka' shows that he fulfilled the role of 'Universal Emperor' (cakravartin). Whereas the Indo-Greeks used basileos basilei (king of kings) title, the Kushanas borrowed titles from the Persians, Chinese and the Romans. They took the titles of maharajadiraja (king of kings), daivaputra (son of heaven), soter (savior) and Kaisara (Caesar).
 - **Hence, option (d) is the correct answer.**

- Q 91. A** • India beat Chinese Taipei 35-28 in the final to claim the Women's Kabaddi World Cup 2025 title in Dhaka, Bangladesh. This was the Indian women's kabaddi team's second World Cup crown on the trot, having also won the inaugural edition hosted in Patna, Bihar, back in 2012. India beat the Islamic Republic of Iran in the final 13 years ago. **Hence, statement 1 is correct.**
- All matches of the second Women's Kabaddi World Cup were held at the Shaheed Suhrawardi Indoor Stadium in Dhaka. **Hence, statement 2 is not correct.**

- Q 92. B** • The Inter-State River Water Disputes are one of the most contentious issues in the Indian federalism today.

Constitutional Provisions:

- Entry 17 of State List deals with water i.e. water supply, irrigation, canal, drainage, embankments, water storage and water power.
- Entry 56 of Union List empowers the Union Government for the regulation and development of inter-state rivers and river valleys to the extent declared by Parliament to be expedient in the public interest.

According to Article 262, in case of disputes relating to waters:

- Parliament may by law provide for the adjudication of any dispute or complaint with respect to the use, distribution or control of the waters of, or in, any inter-State river or river valley.
- Parliament may, by law provide that neither the Supreme Court nor any other court shall exercise jurisdiction in respect of any such dispute or complaint as mentioned above.
- **River Board Act, 1956:** This empowered the GOI to establish Boards for Interstate Rivers and river valleys in consultation with State Governments. Till date, no river board has been created. **Hence statement 1 is not correct.**

The resolution of water dispute is governed by the Inter-State River Water Disputes Act, 1956.

- According to its provisions, if a State Government makes a request regarding any water dispute and the Central Government is of opinion that the water dispute cannot be settled by negotiations, then a Water Disputes Tribunal is constituted for the adjudication of the water dispute. States can not directly approach the Tribunal in any case. **Hence statement 3 is not correct.**
- **The Chairman and members of the Water Dispute Tribunals are nominated by Chief Justice of India. Hence statement 2 is correct.**
- The tribunal has to be constituted within one year of the request.
- The tribunal should give the award within 3 years and in some exceptional cases, within 5 years.
- If the award is not immediately implemented, the concerned parties can seek clarification within three months.

The tribunal award will have the same force as an order or decree of the Supreme Court. The award is final and above the SC's jurisdiction.

- However, the states could still approach SC through Article 136 (Special Leave Petition).
- Private persons could approach the SC under violation of Article 21 (Right to Life).

- Q 93. B**
- The MQ-9B SkyGuardian is developed by General Atomics Aeronautical Systems, which is a company based in the United States (not Israel). It is a High-Altitude Long-Endurance (HALE) drone. **Hence, pair 1 is not correctly matched.**
 - The Hermes-900 is a multi-role Medium-Altitude Long-Endurance (MALE) UAV manufactured by Elbit Systems, which is based in Israel (not the United States). **Hence, pair 2 is not correctly matched.**
 - The Bayraktar TB2 is a famous tactical MALE UAV developed and manufactured by Baykar, a defense company in Turkey. It gained significant global attention during recent conflicts for its surveillance and precision-strike capabilities. It was widely used in conflicts such as Ukraine-Russia war and Nagorno-Karabakh conflict. **Hence, pair 3 is correctly matched.**
 - The Tapas-BH-201 (Tactical Airborne Platform for Aerial Surveillance-Beyond Horizon-201) is an indigenous MALE UAV developed by the Defence Research and Development Organisation (DRDO) of India. Tapas-BH-201 is also known as Rustom-II. **Hence, pair 4 is correctly matched.**

- Q 94. A**
- Hedge funds**, including fund of funds, are unregistered private investment partnerships, funds or pools that may invest and trade in many different markets, strategies and instruments (including securities, non-securities and derivatives) and are not subject to the same regulatory requirements as mutual funds. **Hedge funds:**

- are organized as private investment partnerships or offshore investment corporations;
- use a wide variety of trading strategies involving position-taking in a range of markets;

- employ as assortment of trading techniques and instruments, often including short-selling, derivatives and leverage;
- pay performance fees to their managers; and
- have an investor base comprising wealthy individuals and institutions and relatively high minimum investment limit

Hedge Fund Investment strategies tend to be quite different from those followed by traditional asset managers. Moreover, each fund usually follows its own proprietary strategies which do not always fit within neat definitional categories. Hedge funds could be distinguished by the following three broad types:

- **Event driven** - funds investing in securities to take advantage of price movements generated by corporate events. This group includes merger arbitrage funds and distressed asset funds.
- **Global macro** – funds that take long and short positions in major financial markets based on views influenced by economic trends and events.
- **Market neutral** – funds where the manager attempts to minimize (or significantly reduce) market risk. This category includes long / short equity funds, convertible bonds arbitrage funds, and fixed income arbitrage. **Hence, option (a) is the correct answer.**

Q 95. B • Thus, optio (b) is the correct answer.

Quick Overview of BRICS



Genesis: It has been coined by British economist **Jim O'Neill** in **2001**; BRIC formalized at the **G8 Outreach Summit (2006)**.
 ▶ **1st BRIC Summit** held in **Russia (2009)**.



Members: Original 5: Brazil, Russia, India, China.
 ▶ **South Africa** joined the group in 2010.
New Members: Egypt, Ethiopia, Iran, United Arab Emirates and Indonesia.
 ▶ **Indonesia** has become the first Southeast Asian Member of BRICS.
 ▶ **Saudi Arabia** still had not made a formal decision as of the end of 2024.



BRICS partner countries: Belarus, Bolivia, Cuba, Kazakhstan, Malaysia, Thailand, Uganda, and Uzbekistan, Nigeria.



Secretariat: No permanent secretariat or office.

Q 96. C Köppen's Climate Classification: Dry Climates

- The Köppen Climate Classification is one of the most widely used schemes to categorize the world's climates. It is primarily based on temperature and precipitation. Among its categories, Group B represents Dry Climates, which are defined not by temperature alone but by the relationship between precipitation and evaporation. These regions experience permanent water deficiency, hence vegetation is limited to steppe (grassland) or desert types.

Types of Dry Climates under Köppen

Subtropical Steppe (BSh):

- Semi-arid conditions found on the fringes of subtropical deserts.
- Hot summers, mild winters, with annual rainfall slightly higher than deserts (30–60 cm).
- Example: Margins of the Sahara, Deccan Plateau fringes. **Hence, option 1 is correct.**

Mid-latitude Steppe (BSk):

- Found in continental interiors of mid-latitudes.
- Colder winters compared to subtropical steppe; rainfall modest but insufficient for forests.
- Example: Central Asia, Great Plains of USA. **Hence, option 2 is correct.**

Subtropical Desert (BWh, BWk):

- The driest regions with rainfall less than 25 cm.
- Found under the influence of subtropical high-pressure belts.
- Example: Sahara, Thar Desert, Atacama. **Hence, option 3 is correct.**

Mediterranean (Csa, Csb):

- Not part of Group B (Dry Climates) but classified under Group C (Warm Temperate or Mesothermal Climates).
- Characterized by dry summers and wet winters due to shifting westerlies and subtropical highs.
- Example: Southern Europe, California, Central Chile. **Hence, option 4 is not correct.**

- Q 97. B • The Constitution has not fixed the tenure of a judge of the Supreme Court. Instead it prescribed that he holds office until he attains the age of 65 years. Any question regarding his age is to be determined by such authority and in such manner as provided by Parliament.**
- **The Constitution has not fixed the tenure of a judge of a high court. Instead it prescribed that he holds office until he attains the age of 62 years. Any questions regarding his age is to be decided by the president after consultation with the chief justice of India and the decision of the president is final.**

The Constitution of India (Article 148) provides for an independent office of the Comptroller and Auditor General of India (CAG). He is the head of the Indian Audit and Accounts Department¹. He is the guardian of the public purse and controls the entire financial system of the country at both the levels—the Centre and the state.

- **He holds office for a period of six years or upto the age of 65 years, whichever is earlier.** He can resign any time from his office by addressing the resignation letter to the president. He can also be removed by the president on same grounds and in the same manner as a judge of the Supreme Court.

The Union Public Service Commission (UPSC) is the central recruiting agency in India. It is an independent constitutional body in the sense that it has been directly created by the Constitution. Articles 315 to 323 in Part XIV of the Constitution contain elaborate provisions regarding the composition, appointment and removal of members along with the independence, powers and functions of the UPSC.

- **The chairman and members of the Commission hold office for a term of six years or until they attain the age of 65 years, whichever is earlier.** However, they can relinquish their offices at any time by addressing their resignation to the president. They can also be removed before the expiry of their term by the president in the manner as provided in the Constitution.
- **Hence, option (b) is the correct answer.**

- Q 98. B • The Mother Goddess figurines from the Indus (Harappan) civilisation are characteristically made of terracotta (fired clay). These female figures are among the most common finds from Harappa, Mohenjo-daro and other Harappan sites and are interpreted as fertility/mother-goddess images or domestic votive objects.**



- The majority of Harappan female figurines labelled Mother Goddess are moulded or hand-shaped from clay and then baked i.e., terracotta.
- Such terracotta figurines were inexpensive to produce and are found in large numbers in house-levels and refuse pits, consistent with domestic/folk cult use.
- Bronze: Bronze is the material of the famous Dancing Girl from Mohenjo-daro, not of the ordinary Mother Goddess figurines. Bronze works are rarer and use lost-wax techniques.
- Ivory: Ivory was used for small personal objects and inlays, but not for the widespread Mother Goddess statuettes.
- **Hence, option (b) is the correct answer.**

Q 99. C • The 30th Conference of Parties (COP30) to the United Nations Framework Convention on Climate Change (UNFCCC) is a critical global event hosted in Belém, Brazil, focusing on accelerating climate action and implementing the commitments of the Paris Agreement.

Significance of Belém

- Belém serves as an entry to the Amazon rainforest, a crucial carbon sink and biodiversity reserve.
- The location highlights the urgency of addressing deforestation and land use changes threatening the Amazon.
- **COP30: The 'Implementation COP':** COP30 is dubbed the 'Implementation COP' as it aims to transition from commitments to concrete actions based on the Global Stocktake (GST), assessing progress and identifying gaps in addressing climate change.
- Focus areas include energy, industry, and transport transitions; stewardship of forests, oceans, and biodiversity; transformation of food systems; and resilience in cities and infrastructure.
- **Hence option (c) is the correct answer.**

Q 100. B Bodhisattva in Buddhism

- A Bodhisattva is a person who has developed Bodhicitta (a spontaneous desire and compassionate attitude) in order to achieve Buddhahood for the benefit of all sentient beings.
- It is a Mahayana Buddhist idea that believes in universal emancipation.
- Buddha in his former lifetimes was supposed to be Bodhisattva as contemplated in Jataka tales and the stories portray the different endeavors of the bodhisattva to embrace traits like self-sacrifice and righteousness.

- Prominent Bodhisattvas under Buddhism

Avalokitesvara:

- One of Buddha's three protector deities, also known as Padmapani and is described as carrying a Lotus Flower.
- His picture, which can be found in the Ajanta Caves, is the most well-known of all the Bodhisattvas.
- The compassionate bodhisattva listens to the world's pleas and responds in skillful ways. **Pair 1 is correctly matched**
- Under the name Lokeshvara, he appears informally in Cambodian Theravada Buddhism. He is also shown as a woman, and His Holiness The Dalai Lama is thought to be his incarnation.

Vajrapani:

- Vajrapani is one of three guardian deities that surround Buddha and is represented in the Ajanta Caves.
- He is said to express all of Buddha's powers as well as the powers of the five tathagatas- Vairocana, Akshobhya, Amitabha, Ratnasambhava, and Amoghasiddhi.

Manjusri:

- Manjusri is one of three protecting deities that surround Buddha and is also represented in the Ajanta Caves.
- He is a man Bodhisattva with a sword in his hand who is linked with Buddha's Wisdom.
- Finally, Avalokitesvara embodies Buddha's compassion, Vajrapani embodies Buddha's might, and Manjusri ponders Buddha's knowledge.
- **Samantabhadra:** Linked to meditation and practice. In Buddhism, he is part of the Shakyamuni triad, which also includes the Buddha and Manjusri. **Pair 3 is not correctly matched.**
- **Ksitigarbha:** He is represented as a Buddhist monk who vowed not to attain Buddhahood until hell was totally empty.
- **Maitreya:** A future Buddha who will arrive on Earth, attain perfect enlightenment, and teach the pure dharma. Laughing Buddha is supposed to be an incarnation of Maitreya. **Pair 2 is not correctly matched.**
- **Akasagarbha:** Linked to the concept of space.
- **Tara:** Tara is a Vajrayana Buddhist deity who embodies the values of success in labor and accomplishments.
- **Vasudhara:** He is one who represents riches, prosperity, and abundance. In Nepal, he is quite popular.
- **Skanda:** He is the protector of viharas and Buddhist teachings.

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